



Fall 2025 Industrials Outlook

Manufacturing Space Technology

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Industrials Team



Miles Rucker

Director

720-297-3912

rucker.m@northeastern.edu



Owen Romano

Associate

908-256-1116

romano.o@northeastern.edu



Zachary Cohen

Analyst

646-617-3554

cohen.zach@northeastern.edu



Rohan Kalaria

Analyst

475-731-3865

kalaria.ro@northeastern.edu



Gonçalo Soares

Analyst

973-830-0209

soares.go@northeastern.edu

INTRODUCTION

Current Industrials Landscape

The BBR Team sees sector performance this year being shaped by three forces: the rapid expansion of the AI buildout and its beneficiaries, heightened geopolitical uncertainty and shifting tariff policies under the Trump administration, and broadly supportive economic conditions driven by a more accommodative Federal Reserve. As a result, the industrials landscape is undergoing meaningful transition. Companies are accelerating reshoring and supply-chain restructuring, committing capital to infrastructure upgrades and modernization of aging assets, and tightening their focus on margin management amid sustained cost pressures. Strategic decisions are increasingly influenced by global trade dynamics, while healthier balance sheets and rising shareholder expectations continue to drive elevated interest in M&A. Overall, the sector is adapting to a mix of structural tailwinds and geopolitical volatility that is redefining how industrial firms invest, operate, and compete.

Valuations at an All Time High

Largely, the market is at all time high. P/E ratios have skyrocketed, and investors have seemed to disregard the volatile decision making made at a political scale. In industrials specifically, we are seeing sectors of outperformance, and sectors in perpetual downcycles. The BBR rationalization for high industrials valuations is because of 3 primary macroeconomic tailwinds. Firstly, the big beautiful bill, allowing for a lower corporate tax rate, and therefore higher bottom line margins. Secondly, rate cuts are now being priced in before they even happen. Companies and financial institutions alike are under the impression that if they do not execute on projects, they will fall behind, driving more capex expansion, and an irrelevance to the cash flow statement from a public market perspective. Finally, new depreciation rules allow for companies to depreciate long term assets at a faster rate, meaning that they will ultimately have a stronger bottom line as depreciation as a pre-tax expense, leading to more capex execution in tandem with fear of getting left behind.

Capex Trends

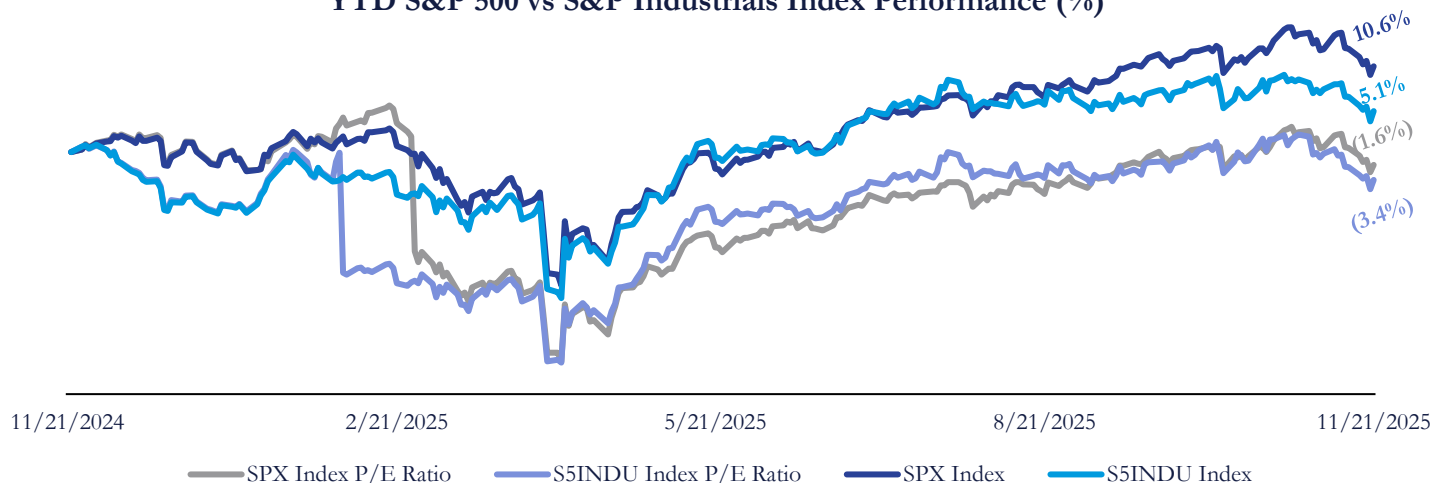
Capex growth across industrials is uneven, with three subsectors seeing the strongest lift from current tailwinds.

Manufacturing (Durable Goods) / Equipment & Structures: Spending is rising as manufacturers upgrade facilities and equipment to support reshoring and higher-value domestic production.

Capital Goods / Automation: Per Morgan Stanley, automation and equipment suppliers are seeing accelerated capex as firms invest in robotics and advanced manufacturing to offset labor constraints.

Infrastructure / Renewables / Utilities: Electrification- and infrastructure-focused companies are benefiting most from the AI datacenter buildout, driving major investment in grid capacity, power systems, and heavy equipment to meet rising demand.

YTD S&P 500 vs S&P Industrials Index Performance (%)

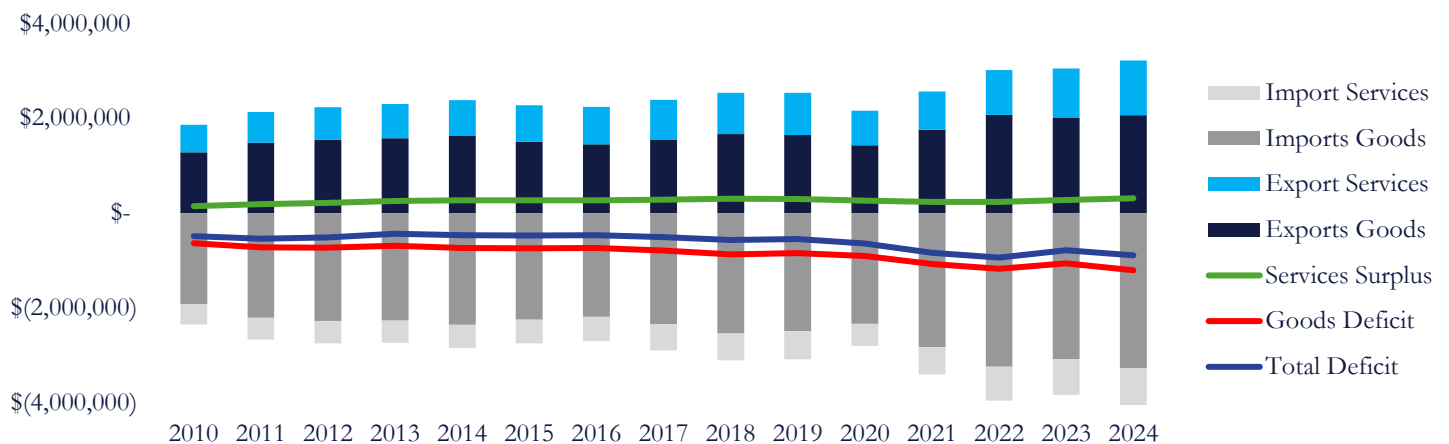


IMPACT OF TARIFFS

China Policy Environment

The U.S. and China are increasingly framed as strategic competitors rather than cooperative partners. Every time our countries seem to come to an agreement, it seemingly is disregarded, and opinions are changed within a couple of days. Looking at it from a historical perspective, we compare this to the space race between the Soviet Union and US, however, this time we are focused on tariffs, and who can win the race to superintelligence. Trade relations remain strenuous and will likely continue to deteriorate under current trajectories. Supply chains are also being reshaped. Not everything is “Made in China” anymore, and American manufacturing reshoring, or offshoring in a more tariff friendly location has been a trend many companies are taking seriously.

US Trade Balance



Trump Tariff Timeline

1Q: Security Revived

March 26 | 25% global tariff on automobiles/certain auto parts

Trump imposed a 25% tariff on imported cars & trucks effective April 3, and specified parts effect May 3

April 2, 2025 | “Reciprocal tariff” framework EO

Liberation Day: reciprocal tariffs program to address partners with persistent goods trade surpluses with the US

2Q: Acceleration

June 3, 2025 | Steel and aluminum raised to 50% (Section 232)

Proclamation increases prior Section 232 tariff rates on steel and aluminum articles and derivatives

June 16 & 30, 2025 | Implementation carve-outs and UK TRQ detail

Tariff rate quotas and part classifications announced, including treatment for UK autos/parts under bi-lateral deal

3Q: Industry Shielding

July 30, 2025 | Copper: 50% Tariff on specified semi-finished copper products

Imposing 50% tariff effective August 1 on certain copper products (excludes ores, concentrates, cathodes, & anodes)

September 30, 2025 | Timber, lumber, and wood furniture tariffs

Announcement of tariffs including 10% on softwood lumber and higher rates on cabinets & wood furniture

4Q: Market Discipline

October 8-22, 2025 | Administration of auto-tariff programs

Commerce opens a new process tied to the March auto proclamation for parts-scope management

October 26, 2025 | U.S. – China trying to avert 100% across the board tariffs

Reporting describes a provisional understanding to avoid threatened 100% tariffs and to address rare earths/agricultural trade

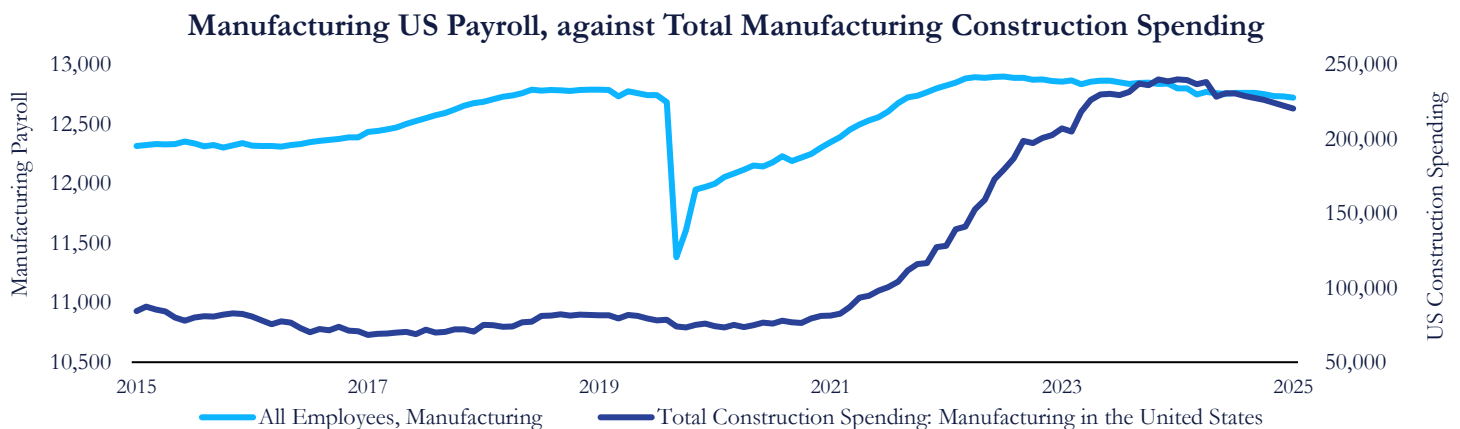
MACROECONOMICS

Manufacturing PMI (2Y)



Is the Industrials Economy Healthy?

From a broad perspective, the industrials economy is *soft* rather than *healthy* right now. Output is inching up, but manufacturing activity and employment are under pressure, with tariffs and uncertainty weighing on orders. As of August 2025, U.S. manufacturing employment is down ~78k YoY. August alone was ~12k jobs. Supporting this tough job market, manufacturing job openings were also down to ~409k in August, from ~438k in July. As an economy, there is a clear discrepancy. Demand for the jobs is easing, and total jobs are also decreasing. Firms are firing, and not hiring back, one of the downsides of the expectations from AI efficiencies. U.S. Industrial production (Manufacturing, mining, utilities) was up about 0.9% YoY in August 2025. In comparison to other industries, this is relatively weak. Another important statistic to look at is the ISM Manufacturing PMI. In October, at 48.7, it was the 8th straight month in contraction (<50), driven by weak new orders and softer production.

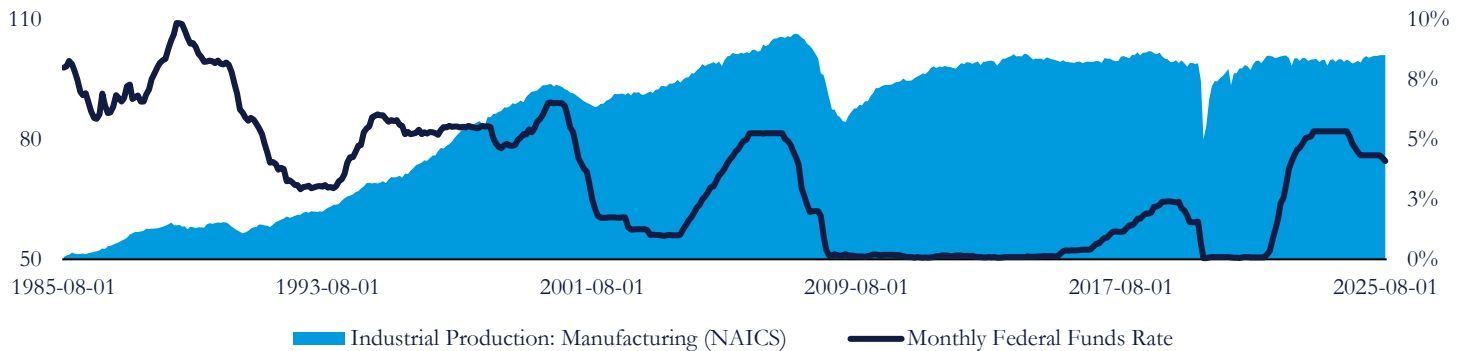


Correlation: Weaker Industrials Labor Market & Lack of Health

The fact that the manufacturing workforce is shrinking, signals weak demand, order deferrals, or cost pressures limiting new hiring. A decline in hours worked (even if modest) suggests that firms are rightening labor usage rather than adding capacity, which is consistent with cautious capex/spending behavior. As mentioned above, the drop of ~78K jobs YoY is meaningful for a goods-producing sector that typically leads in economic expansions. On the flip side of the weak statistics, the IMF projects global growth around 3.2% in 2025. When economies grow, especially in developing markets, a strong industrials backdrop is a necessity to facilitate said growth. Looking at the biggest market, being the U.S., the positive sentiment is not the same. Policy is a big swing factor: Tariffs are a tax on the industrials system, even while they're meant to support domestic production. All said, the industrial economy is in a "soft patch", not in a crisis, but clearly not healthy. It's currently characterized by weak manufacturing sentiment, modest real output growth, easing labor demand, and policy-driven uncertainty.

HISTORY OF MONETARY POLICY

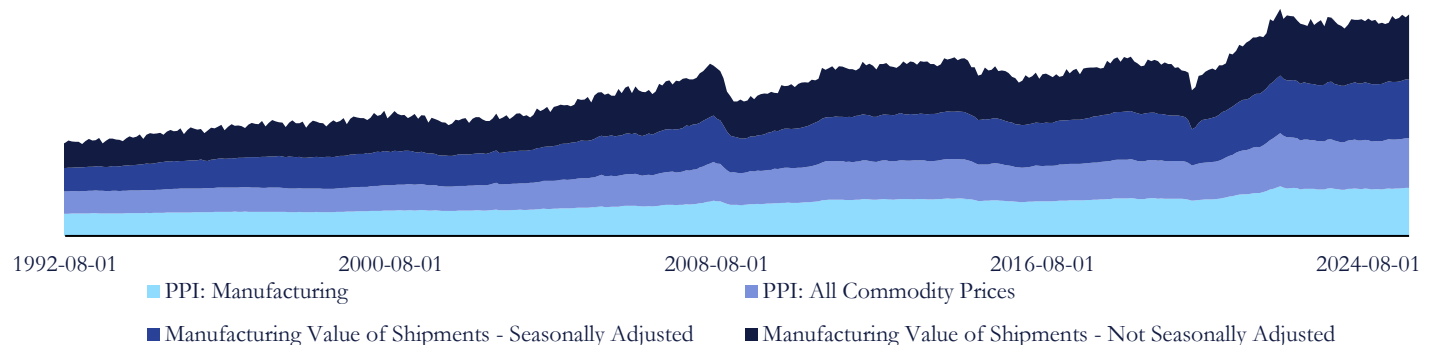
Fed Rates/Industrials Production Timeline: 40 Years



Where We Are Headed Now

Macroeconomic statistics are weak: Payroll is down, companies are delaying their spending processes, tariffs are clouding the water, and the FED is inconsistent, but correct in its sentiment. While U.S. industrials currently remain weak, our view is that we will see a point of overall inflection starting in the first half of 2026, backed by political clarity, Big Beautiful Bill depreciation implications, a lower corporate tax rate, a consistently dovish FED with data to support, and therefore firms that want to hire employees backed by the desire of companies to execute and finalize orders on projects. Industrials remains the backbone, and foundation to U.S. security, the AI Buildout, onshoring of U.S. manufacturing, and navigation of logistics and political instability. Many are predicting an inflection of the overall U.S. economy in 1/2Q of 2026, with the tailwinds mentioned above. If these things come to fruition, the industrials economy will be the first noticeable places where statistics and valuations normalize.

Inflation & Input Costs

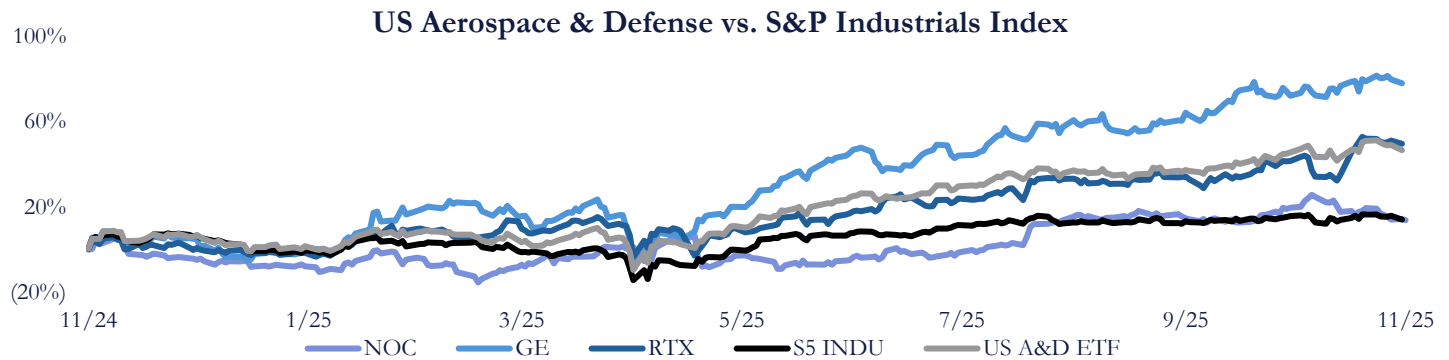


Dual Mandate Could Hit Industrials Hard

As of late, our Federal Reserve has made it a point to emphasize labor statistics, over inflation statistics. This has led our inflation to miss its target for years. If corporations are continuously in a battle for more pricing power, it hits industrials especially hard. Industrials have 3 main considerations: Commodities/Materials, Labor Inflation, and Energy/Logistics. On the commodities and materials side of things, when input prices rise, gross margins compress if companies cannot raise selling prices. Firms with longer lead time contracts face a lag in passing through those increases. For the labor side of things, average hourly earnings are up roughly 4.1% Y/Y. With everyday items such as food, housing, and utilities increasing, unionized or skilled workers continuously demand higher wages, adding to the perpetual cycle of sticky inflation. Finally, energy inflation filters through logistics and plant operations. For transportation and aerospace subsectors, high fuel costs directly increase operating expense and indirectly hit demand. Inflation needs to get under control if the industrials economy is to organically thrive.

PERFORMANCE ANALYSIS

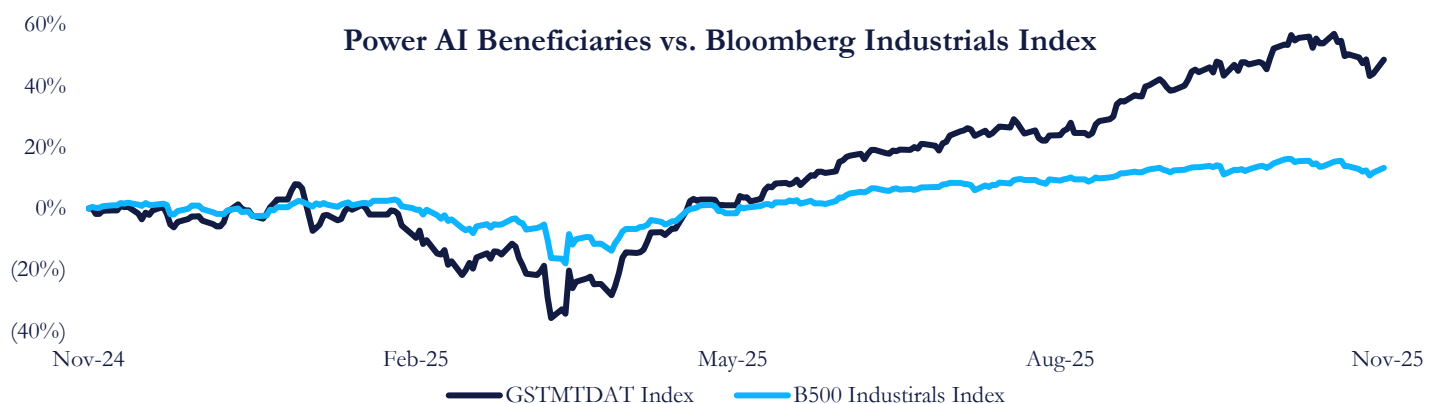
Aerospace & Defense



Our Perspective

Aerospace and Defense continues to outperform the broader Industrials index, supported by structural demand for national security amidst geopolitical tensions, the modernization of defense systems, and increasing government funding allocations. We believe that this outperformance is durable. Defense spending is not only increasing but becoming less cyclical, commercial aerospace continues to resurge post-pandemic, and firms which serve both commercial and government clients will keep benefitting from higher-margin aftermarket and service revenue, along with long-term contract visibility. As budgets continue shifting to prioritize space domain awareness, hypersonics, and autonomous systems, we expect A&D to remain one of the most resilient and attractive segments within broader Industrials. Those who can convert AI adoption, nearshoring, and policy support into structural cost advantages and higher-margin service revenue, will see substantial upside.

Electrical and Power Infrastructure Underperforms



AI Beneficiaries

AI integration across industrial processes is emerging as the next structural productivity cycle, as it is linking automation, data analytics, and operational efficiency. The global industrial AI market is expanding rapidly, supported by growing demand for predictive maintenance and process optimization tools. Leading automation companies such as Rockwell Automation, Siemens, Emerson, and Honeywell are embedding AI capabilities into control systems and edge computing platforms to enhance uptime, reduce energy consumption, and grow recurring software revenue. Industry research suggests that AI-driven automation could lift manufacturing productivity by up to 20%, while incremental efficiency gains may contribute modest margin expansion across diversified industrial technology firms. As adoption scales, investors are rewarding early movers with significantly higher valuation multiples relative to legacy automation peers, reflecting long-term confidence in durable, data-driven earnings growth.

THE BBR VIEW

Where Industrials is Heading

Our team sees the industrials sector underperforming in 2026 due to persistent inflation, elevated commodity costs, and muted consumer confidence, which together limit pricing power and slow demand across core sub-industries reasons. A still-uncertain geopolitical backdrop and cautious corporate capex further constrain growth prospects, particularly for manufacturing and transportation names. However, industrials remain the foundation of the U.S. economy, and if a more dovish Fed can stabilize inflation and restore visibility on rates, we could see an inflection point with re-acceleration in capex, reshoring initiatives, and supply-chain reinvestment. The health of the sector is fundamentally consumer-driven, and if inflation stays sticky, margins will remain under pressure.

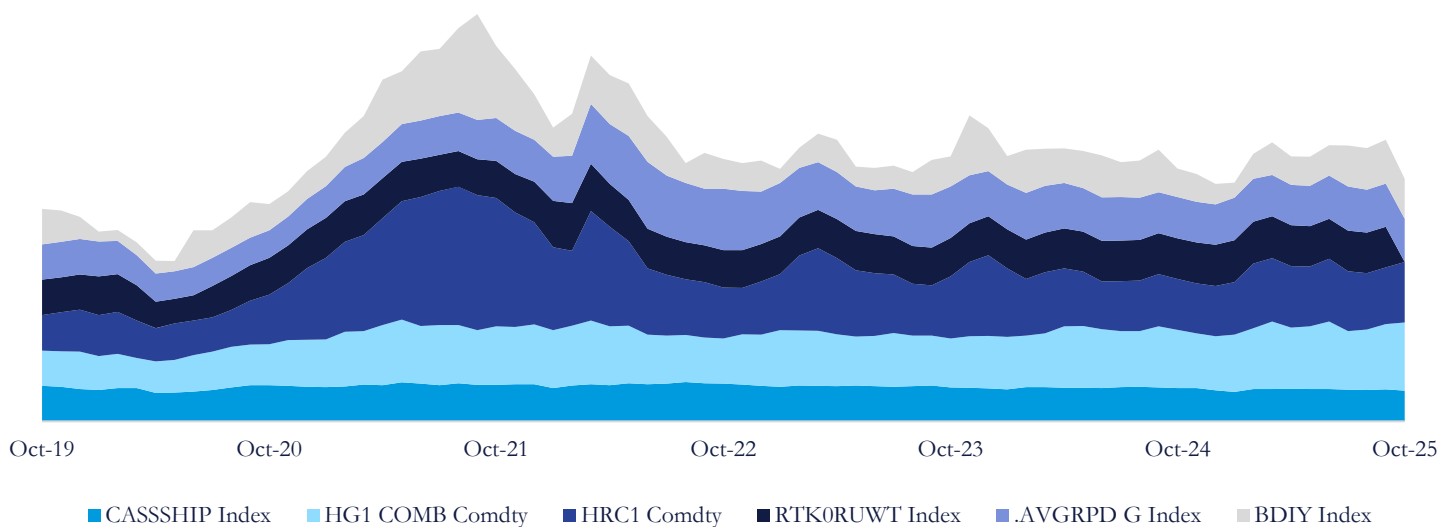
Tailwinds

There are several factors that lead to optimism within in the industrials sector and could be seen as tailwinds. Due to geopolitical risk and advantageous government policies, many companies are relocating production to domestic sites from overseas. As many developed economies worldwide have faced underinvestment in infrastructure, there are sizable opportunities and large-scale projects to undergo, leading to business in the industrials sector. Industrial firms are also heavily investing in digital manufacturing, AI, and other technology-based operations to boost their outputs and become more efficient. As a greater emphasis on climate change and sustainability becomes mainstream, numerous companies that have clean technology capabilities are set to prosper. Lastly, the COVID-19 Pandemic had a large impact on the sector and as we get further and further away from it, companies that faced supply-chain disruptions or inventory gluts no longer have such challenges.

Headwinds

The industrial sector is challenged by rising input and energy costs, which continue to put pressure on margins and operational performance. Prices for steel, copper, and other critical materials remain elevated, while energy costs fluctuate with global markets, making production planning and budgeting more complex. These cost pressures are compounded by the limited ability to fully pass them through to customers in competitive markets, leaving many firms financially constrained. At the same time, labor shortages represent a strategic challenge. Skilled positions are increasingly difficult to fill, driving wage inflation and slowing production across key processes. Together, these two challenges create a backdrop of persistent headwinds that can influence strategic decisions across investment, capacity planning, and technological adoption. They underscore the importance of operational efficiency, supply chain resilience, and workforce development in maintaining long-term competitiveness.

Market Perspective: A Mix of Industrial Economic Indicators



A photograph of the Space Shuttle Discovery launching from the launch pad. The shuttle is white with a large orange external tank and white solid rocket boosters. It is ascending vertically, leaving a large plume of white smoke and fire from its engines. To the left of the shuttle is the yellow service structure of the launch pad. The sky is blue with some light clouds.

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Industrials M&A Activity

Union Pacific (UNP) acquisition of Norfolk Southern Corp (NSC)

Transaction Overview

UNP announced its acquisition of NSC for ~\$85B in equity value, and a new combined EV > \$250B. Union Pacific acquisition marks a significant milestone, and with potential for implications at global logistic scale, with a new coast to coast U.S. freight railroad.

Strategic Rationale

Creates single-line transcontinental services vs current handoffs between west/east carriers. Expected to cut coast-to-coast transit times by up to ~2 days, especially by reducing interchange congestion in popular hubs.

Norfolk Southern Company Snapshot

Class 1 freight railroad focused on the eastern U.S. Currently has around 19.5K of route miles, servicing every major eastern container port. NSC currently has strength in intermodal freight, and ~800 industrial sites, 175 warehouses, and 315 transload locations.

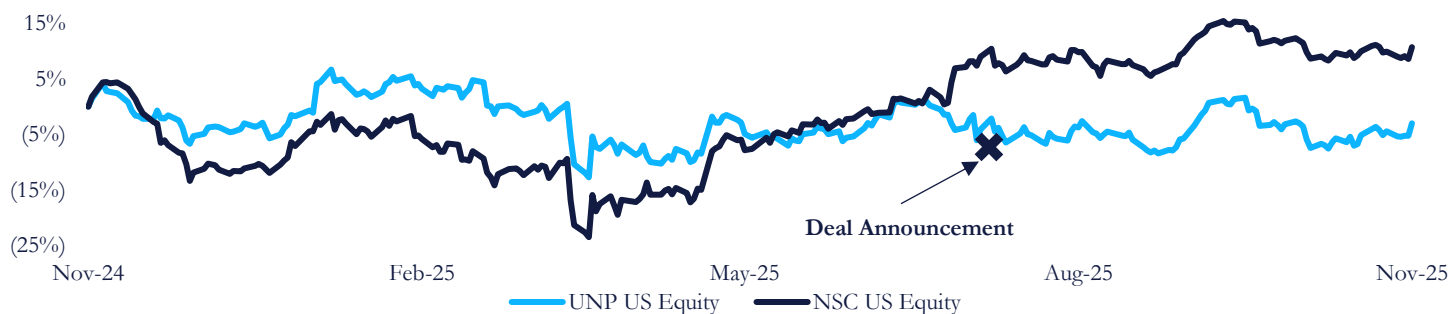
Impact on Acquirer

Moves UNP from west-focused franchise to a national rail platform. UNP suspended buyback plan until 2028 but maintains dividend. Additionally, they anticipate \$2.75B in synergies, and EPS accretion in Year 2.

Timeline & Risks to Consider

Deal was announced on July 29th, 2025, and has a target close of Early 2027. Regulatory challenges lie ahead, with competition essentially being cut out, and unions and advocacy groups already warning and rallying against job cuts, service degradation, and higher derailment risk.

UNP vs. NSC Stock Price YoY (Normalized)



Deal Terms

Transaction Type	Stock + Cash M&A of NSC by UNP
Price per Share	\$320 per NSC share
Equity Value	~\$72 billion
Enterprise Value	~\$85 billion
Financing	Cash funded through debt/BS; 225M new shares
Expected Close	Early 2027

UNP Info

UNP	2022A	2023A	2024A
Revenue	\$24.9B	\$24.1B	\$24.3B
Operating Margin	39.8%	37.7%	40.0%
Profit Margin	28.1%	26.5%	27.8%

NSC info

NSC	2022A	2023A	2024A
Revenue	\$12.7B	\$12.2B	\$12.1B
Operating Margin	37.8%	23.8%	33.9%
Profit Margin	25.7%	15.0%	21.6%

UNP Stock Info

Market Cap (B)	\$131.1
Shares Outs. (mm)	624
Free Float	99.7%
Dividend Yield	2.5%

NSC Stock Info

Market Cap (B)	\$63.2
Shares Outs. (mm)	236
Free Float	99.9%
Dividend Yield	1.9%

Baker Hughes (BKR) acquisition of Chart Industries (GTLS)

Transaction Overview

BKR announced its acquisition of GTLS through an all-cash transaction on July 29, 2025. The deal marks significant consolidation in the energy equipment sector and is subject to standard regulatory and shareholder approvals.

Strategic Rationale

BKR intends to expand its energy tech strategy with GTLS, boosting Middle East/Asia presence, and doubling non-oil/gas exposure, targeting \$325M annual synergies and additional revenue from cross-selling.

Chart Industries Company Snapshot

GTLS operates in the industrial gas and clean energy equipment sector, providing cryogenic solutions for LNG, hydrogen, and other gases. It has strong positions across LNG, industrial, and power markets, with rising exposure to specialty gases like hydrogen and helium.

Impact on Acquirer

The deal is expected to deliver double-digit EPS accretion in the first year and double-digit ROIC by Year 5, with leverage falling from 2.25x to 1.0x-1.5x within two years.

Timeline & Risks to Consider

Next steps include securing antitrust and global regulatory approvals, achieving full \$325M synergies by Year 3, and de-leveraging within 24 months. Key risks include cultural integration and talent retention, potential antitrust delays, and higher leverage (~\$13B liquidity) if cash flows weaken amid economic or competitive pressures.

BKR vs. GTLS Stock Price YoY (Normalized)



Deal Terms

Transaction Type	All-cash acquisition
Price per Share	\$210 (22% premium to pre-announcement)
Equity Value	~\$9.5B (based on 45M shares outstanding)
Enterprise Value	~\$13.6B (~9x 2025E EBITDA)
Financing	Fully funded via cash and debt
Expected Close	Mid-2026 (subject to change)

BKR Info

UNP	2022A	2023A	2024A
Revenue	\$21.8B	\$25.5B	\$27.8B
Operating Margin	5.6%	9.1%	11.1%
Profit Margin	20.8%	20.6%	21.0%

GTLS Info

NSC	2022A	2023A	2024A
Revenue	\$1.6B	\$3.4B	\$4.2B
Operating Margin	9.4%	11.7%	15.6%
Profit Margin	5.1%	1.7%	5.7%

BKR Stock Info

Market Cap (B)	\$47.0
Shares Outs. (mm)	997
Free Float	99.8%
Dividend Yield	2.0%

GTLS Stock Info

Market Cap (B)	\$9.40
Shares Outs. (mm)	45
Free Float	99.4%
Dividend Yield	NA

Thoma Bravo acquisition of Digital Aviation Solutions/Boeing Co (BA)

Transaction Overview

Boeing's sale of its DAS unit reshapes its portfolio, carving out its aviation software business into a major standalone platform. The deal positions Thoma Bravo to scale a deeply embedded, high margin ecosystem across global flight operations.

Strategic Rationale

For Thoma Bravo, the acquisition provides a scaled, mission-critical aviation software platform with recurring revenue, strong customer retention, and room for operational and commercial expansion.

Chart Industries Company Snapshot

Digital Aviation Solutions is a low-churn, safety-critical business providing mission-critical aviation software and data for flight operations, serving airlines, business aviation, and pilots through key brands including Jeppesen, ForeFlight, and AerData.

Impact on Acquirer

The deal is expected to be accretive given DAS' strong recurring revenue. Integration requires separating systems and ~4,000 employees while maintaining customers, with success reliant on talent retention.

Timeline & Risks to Consider

Next steps include securing regulatory approvals, completing the carve-out, and establishing Thoma Bravo's standalone platform. Key risks involve execution challenges, regulatory scrutiny over safety-critical data, and customer or talent loss during integration.

Boeing Stock Price YoY



Deal Terms

Transaction Type	All-Cash Carve Out acquisition
Price per Share	Undisclosed
Equity Value	~\$6 billion
Enterprise Value	~\$10.55 billion
Financing	Private-loan financing led by Apollo
Expected Close	End of 2025 (subject to change)

Digital Aviation Info

UNP	2022A	2023A	2024A
Revenue	\$1.7B	\$1.9B	\$2.0B
Operating Margin	25.7%	26.1%	26.5%
Profit Margin	16.8%	17.1%	17.6%

Comparable Transactions

M&A Deal	Premium	EV/EBITDA	P/E
Carlyle/ Standard Aero	25%	14.8x	--
Thales/ Cobham Aerospace	22%	15.5x	--
Honeywell/ Civitanavi	20%	16.2x	28x

BA Stock Info

Market Cap (B)	\$152.8	Free Float	82.0%
Shares Outs. (mm)	756	Dividend Yield	NA

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Public Market Coverage



SPACE TECH MANUFACTURING

Manufacturing Overview

The Space Manufacturing industry encompasses the design, engineering, and production of hardware and technology that enable modern space exploration, satellite communication, and national defense. Core segments include launch systems, satellite components, ground control systems, and in-orbit infrastructure, all serving as the backbone of both government and commercial space initiatives.

The industry’s overarching objective is to advance global connectivity, strengthen defense capabilities, and expand scientific frontiers through sustained innovation in aerospace engineering. As public agencies and private firms accelerate investment in satellite networks, reusable launch vehicles, and space-based data services, space manufacturing continues to evolve as a critical enabler of the next generation of communication and security infrastructure.

Key Terms: Space Tech Manufacturing

Miniaturization & Modularity: enable smaller and easily reconfigurable space systems that lower costs and enhance mission flexibility

GPU: specialized chips designed for high-speed processing that power AI systems crucial to modern space technology

Launch Vehicle: Rocket systems that transport payloads into orbit

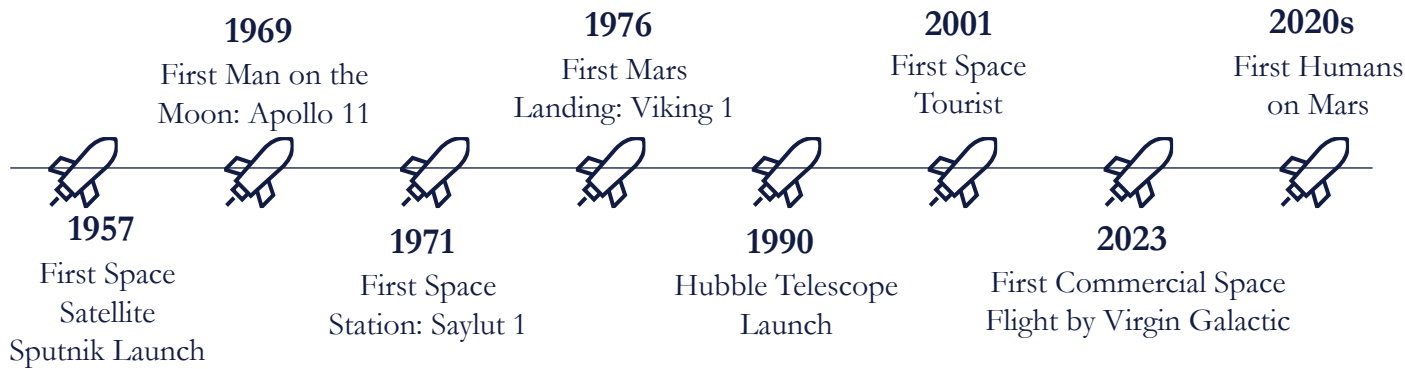
Payload: The cargo carried by a launch vehicle: satellites, instruments, or crewed modules

Constellation: A network of satellites working together for global coverage

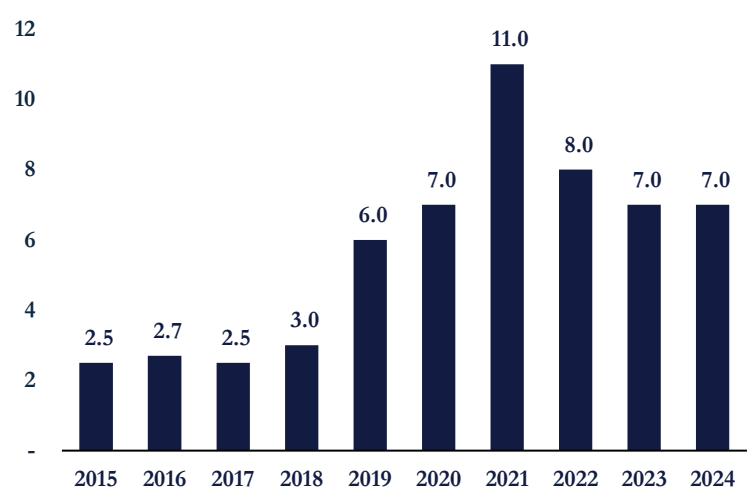
CubeSat: Miniaturized, cost-efficient satellites used for research and commercial applications

LEO (Low Earth Orbit): Orbit up to 2,000 km altitude main operating zone for communication satellites.

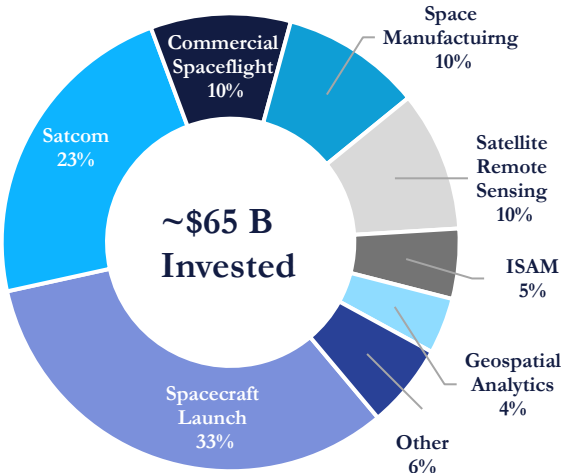
Human Experience in Space



Total Value VC Investments in SpaceTech Startups (\$ Billions)



Deals Completed By Primary Market



Sources: PwC Report, SpaceNews

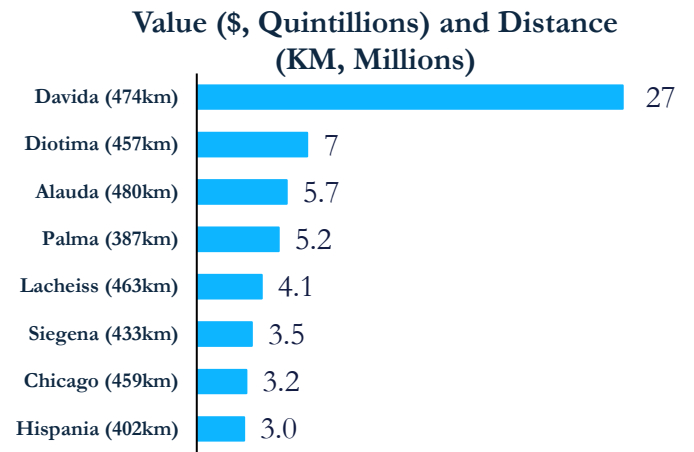
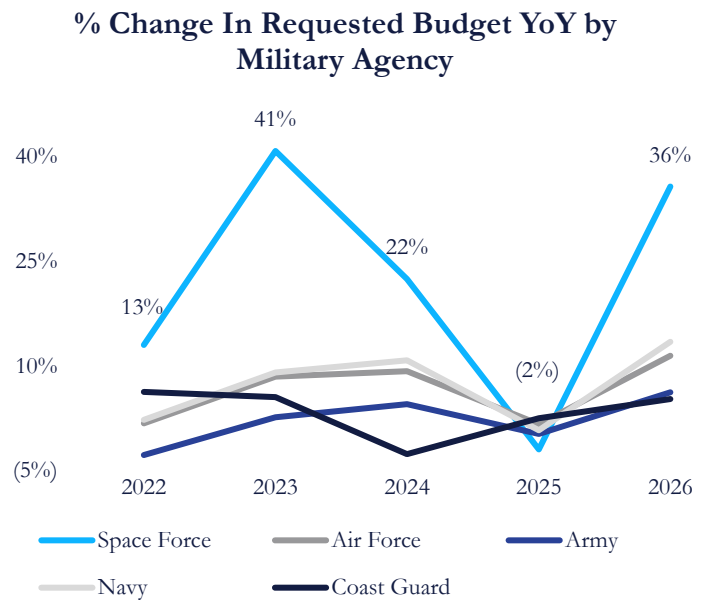
MARKET OVERVIEW

Miniaturization Enabling

The miniaturization of satellite technology is reshaping the economics of space exploration by sharply reducing costs and lowering barriers to entry. Smaller, lighter systems decrease launch mass and fuel demand, enabling more efficient, cost-effective missions and supporting higher launch frequency with greater mission flexibility. This shift has broadened access to space for commercial, academic, and government operators, creating new opportunities in deep-space exploration, lunar payload delivery, and large-scale distributed constellations.

Rising Government Investments

The U.S. policy environment has turned increasingly supportive of commercial space activity, following an Executive Order signed in August 2025 that streamlines environmental approvals, modernizes spacecraft launch regulations, and enhances coordination across federal spaceport infrastructure. This regulatory momentum coincides with a sharp increase in federal spending, including a \$2 billion expansion of NASA’s FY2025 budget, additional FAA funding for Advanced Air Mobility (AAM) integration, and over \$500 million of funding committed to Low Earth Orbit (LEO) commercialization through private space station initiatives. Together, these policy and funding actions signal a coordinated national effort to accelerate public-private collaboration in space manufacturing, launch services, and orbital infrastructure. Long-term projections underscore the sector’s potential, with PwC forecasting total industry value approaching \$2 trillion by 2040, driven by sustained innovation in satellite networks, in-orbit servicing, and reusable launch technologies.



Emerging Opportunities

Emerging commercialization themes in the space sector extend well beyond traditional launch and satellite services. Lunar resource extraction, space-based energy generation, and in-orbit construction and manufacturing are rapidly evolving as strategic frontiers, attracting private capital and government partnership initiatives. In addition, asteroid mining is rapidly shifting from science fiction to a realistic economic frontier. As launch costs fall and space technology accelerates, private firms and national agencies are looking beyond our planet for critical resources. Instead of relying on finite reserves, we can tap untouched celestial bodies that hold massive quantities of valuable metals. The expansion of launch rideshare and “space trucking” services has significantly lowered the cost of access to orbit, shifting industry focus from rocket engineering to payload delivery and enabling more companies to deploy satellites, research instruments, and cargo in cislunar space. At the same time, the communications landscape is transforming as major players invest heavily in next-generation satellite connectivity, most notably Starlink’s \$18 billion spectrum acquisition, underscoring the long-term value of orbital bandwidth as a critical infrastructure asset.

REGULATORY LANDSCAPE

Certifications and Air Mobility

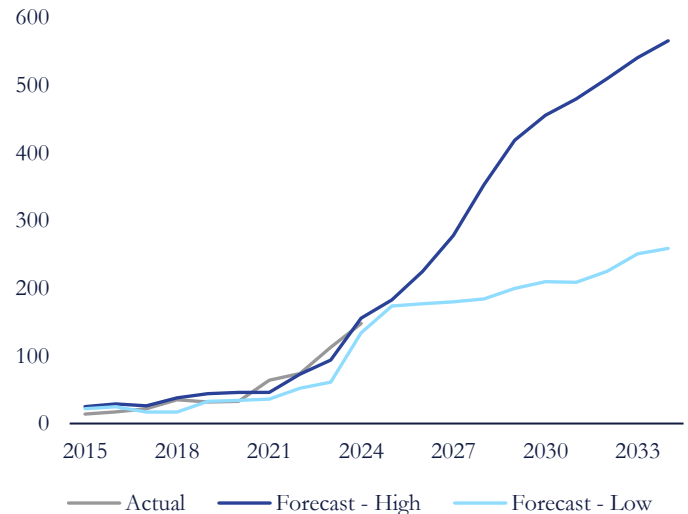
The Federal Aviation Administration (FAA) is the main gatekeeper and primary body that oversees and governs space activity in the United States, from commercial launches to vehicle certification to licensing for reentry. Recently, the FAA has almost become a bottleneck, slowing down launch activity. Similarly, aviation/space regulators are tightening alignment, the European Union Aviation Safety Agency (EASA) and UK Civil Aviation Authority (CAA) have started to establish their own certification frameworks, often copying US best practices.

Infrastructure and Exploration

Much of the regulatory environment regarding lunar and deep-space exploration revolves around Artemis and CLPS (Commercial Lunar Payload Services) – two NASA programs that provide funding and compliance benchmarks that companies must abide by to participate in the U.S. return-to-the-Moon effort. These companies, such as Intuitive Machines (LUNR) and Astrobotic rely on NASA and FCC approvals to best maintain liquidity and to continue to progress. Additionally, there has been an increase in international collaboration as there has been growing regulatory alignment and coordination between agencies such as EASA, JAXA, and the UAE Space Agency. Europe especially has seen a large resurgence through strategic sovereignty concerns and emerging industrial policies such as the European Commission’s IRIS constellation, a €6billion public-private partnership led by Airbus, Eutelsat, and SES, designed to create secure, sovereign communications for defense/government applications.

On the investor side, such programs help to create a more stable long-term regulatory outlook and creates a broader base of potential companies that could get involved. Overall, regulatory milestones do often serve as regulatory catalysts and beyond just launch services there is an increasing number of space infrastructure companies, showing this space is one with a good amount of potential growth.

FAA Commercial Launches Per Year

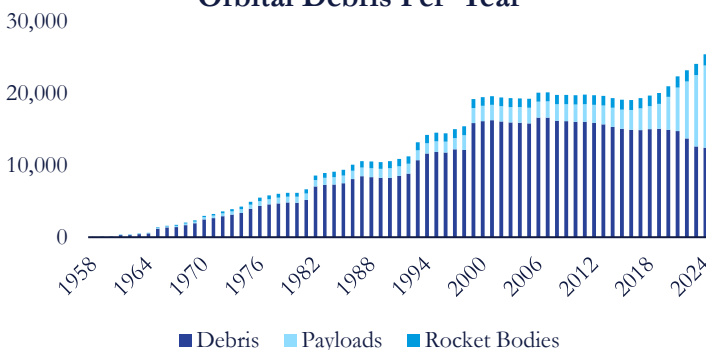


Spectrum and Sustainability

Spectrum management and orbital sustainability have become increasingly important as satellite constellations become more common. Agencies such as the U.S. Office of Space Commerce, the FCC, and International Telecommunication Union have started to focus on preventing frequency interference and congestion as well as developing debris removal and satellite deorbit guidelines. On frequency interference, these agencies must focus on players such as Starlink, OneWeb and AST SpaceMobile. Satellite broadband constellations can strain limited radio-frequency bands, and frequency coordination has a direct influence on deployment timelines and service reliability.

Orbital debris mitigation has become formal regulation with agencies requiring end-of-life deorbit plans. This is often backed by ESG frameworks which can demand operators to disclose sustainability and safety metrics. Companies with a strong record in debris mitigation and compliance may have lower regulatory risk and therefore be more attractive for investment.

Orbital Debris Per Year



SPACE-BASED DATA CENTERS

Why Space?

Space offers a near-zero-temperature environment (~ 3 K background) enabling passive radiative cooling. This could help mitigate the current energy dedicated to datacenters (cooling currently accounts for 30-40%). Energy Autonomy is another tailwind for this. Uninterrupted solar in specific orbits (e.g., GEO) enables continuous solar power generation, eliminating intermittency and reducing reliance on terrestrial grids. Finally, there is physical security & sovereignty in space. With this, there is immunity from terrestrial outages, weather events, & geopolitics.

Industrial & Technological Feasibility

Launch Cost Trends: The decline in cost/kg (SpaceX, Rocket Lab, Relativity) makes deployment economically plausible

Satellite Platform Scalability: Reuse of LEO and MEO bus architectures for modular compute/storage payloads

Power generation: Continuous solar exposure in geosynchronous orbits could enable energy-independent systems

Thermal management advantage: Vacuum of space offers efficient passive heat dissipation (key for cooling GPUs/TPUs)

Challenges: Radiation shielding, maintenance logistics, data latency for Earth-bound users, debris risk, and secure data transmission

Risks & Barriers

There is a clear latency issue, with GEO introduces ~ 600 - 700 ms round-trip latency. Only certain workloads are suitable with this. Space isn't completely immune from "weather", as cosmic rays and solar storms could degrade electronics and memory. Finally, robotics are not at a stage currently to support the datacenter maintenance.

The Industrial Value Chain



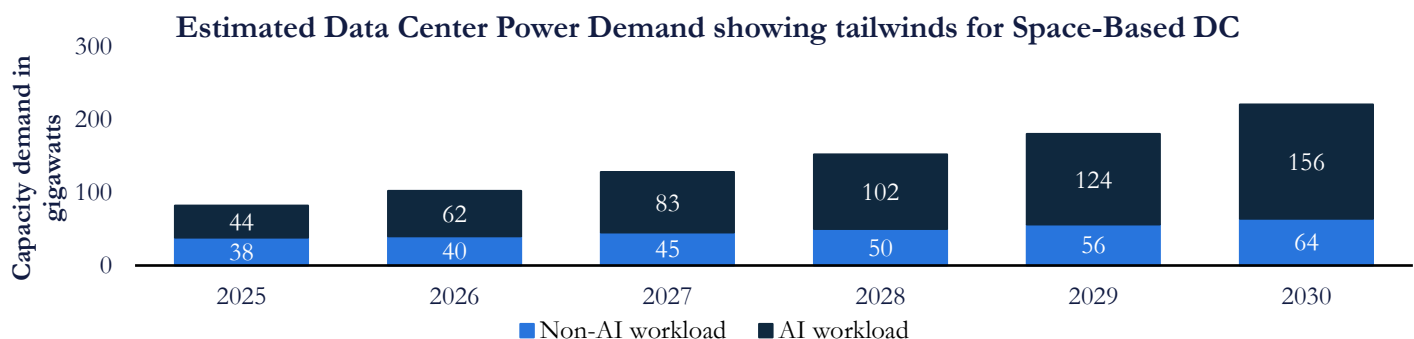
Newly Renamed Starcloud raises \$10M for space data centers



Economic Case

Drivers of Capital Expenditure: Launch services are priced per kilogram, orbital insertion and station-keeping propulsion, and specialized radiation-hardened components.

Drivers of Operating Expenditures: Insignificant cooling expenses and data infrastructure management.



RECREATIONAL SPACE EXPLORATION

Sector Overview

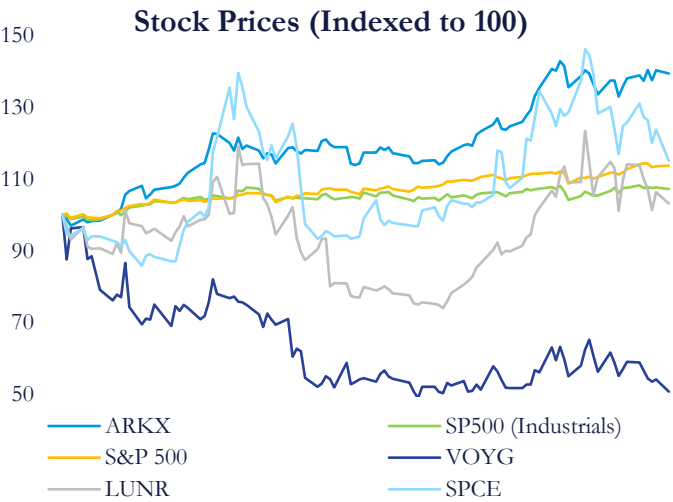
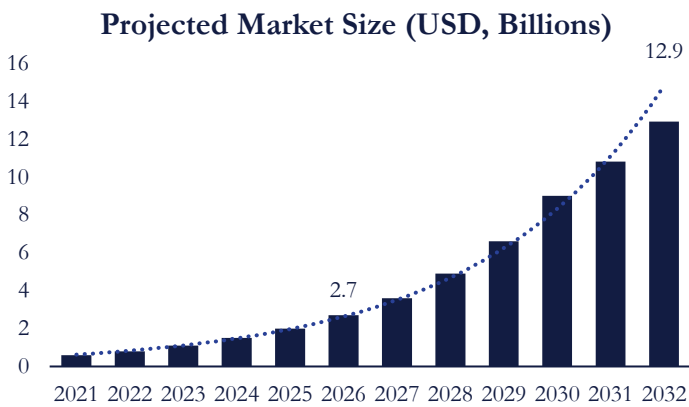
Recreational space exploration is the consumer-facing edge of the space economy. This sector encompasses both suborbital and orbital human travel, microgravity experiences, and long-duration habitation missions. Historically driven through state programs, the sector is transitioning towards commercial and private entrants. The overall sector focuses on human experience, tourism economics, and consumer safety certifications. The TAM is estimated to be ~\$8B–\$12B, with growth reliant on launch costs, flight frequency, and infrastructure development.

Competitive Landscape

The recreational space exploration market is highly fragmented and concentrated, with early entrants largely spanning three categories: suborbital tourism, orbital infrastructure providers, and lunar enablers. Most of the industry is through private companies, but there is public-market exposure through companies such as Virgin Galactic (SPCE), Voyager Technologies (VOYG), and Intuitive Machines (LUNR). Three companies that all revolve around different segments within recreational space exploration.

Virgin Galactic (NYSE: SPCE)

Virgin Galactic is the only publicly traded pure-play company that provides exposure to suborbital space tourism. Their SpaceShipTwo platform, a reusable space plane that carries passengers ~50 miles above the earth surface for low gravity exposure, is what Virgin is most known for. Their business model is reliant on high-fixed costs and low volume, with tickets often starting at ~\$450,000. The company has also run into many operational challenges, such as FAA re-certification delays, gaps between flights, trouble with scaling, and cash burn; leading to waning investor confidence. Management decided to make a strategic pivot to focus on Delta-class spacecraft, more robust manufacturing cadences with higher flight frequency and lower ticket costs, expected to enter service in late 2026.



Voyager Technologies (NYSE: VOYG)

Voyager Technologies represents the infrastructure part of the recreational space exploration segment. Voyager’s model is vertically integrated – using orbital infrastructure, life support systems, and in-space manufacturing. Their flagship initiative is the Starlab commercial space station, being developed in partnership with Airbus and NASA’s LEO program. More so, Voyager sets themselves apart through their work with governments and scalability, shown through their role in the IRIS program.

Intuitive Machines (NASDAQ: LUNR)

Intuitive Machines operates at the intersection of lunar exploration and commercialization, ultimately providing technology used in government and private lunar methods. A leading contractor in NASA’s CLPS program, LUNR is responsible for developing and operating lunar landers, communications relays, and navigation networks, all which are vital for sustained human presence on the moon. LUNR’s IM-1 mission (2024) was the first private spacecraft to land on the moon establishing them as a credible operator. LUNR is less developed than other companies, with revenues currently dominated by NASA contracts.

M&A ACTIVITY

Firefly Aerospace Acquisition of SciTec

11/05/2025

\$855 Million

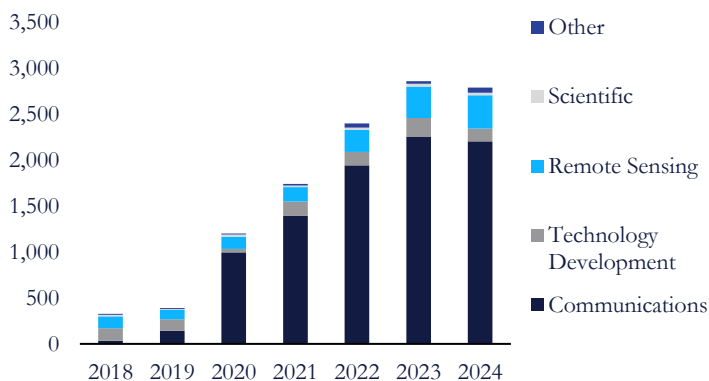
Firefly Aerospace

SciTec

In November 2025, Firefly Aerospace closed the acquisition of SciTec Inc. The transaction comprised of \$300 million in cash and \$555 million in Firefly shares. Firefly Aerospace is an American space and defense company that specializes in launch vehicles that are capable of operations both through orbit and landings. The firm boasts its short turnaround times, being the only commercial company to launch a satellite to orbit within a 24-hour notice period. SciTec is an American technology and software provider with a focus on defense tech ranging from command systems to missile defense systems.

The acquisition deeply enhances Firefly's service offerings, allowing them to continually operate their technologies following their launches. This comes at a time where American investment in missile defense, and space awareness programs are receiving increased allocation. In January 2025, an executive order from The White House described a "Golden Dome" project focusing on American defense from foreign missile threats. The proposed infrastructure of the project is largely centered around the space sector. Following this, a September 2025 report described a \$728 million investment in space operations capability. The newly vertically integrated Firefly Aerospace is well-positioned to benefit from this focus, especially given their history of contracts with the U.S. DoD.

SmallSat Launches by Mission Type



KBR, Inc. Acquisition of LinQuest Corporation

09/03/2024

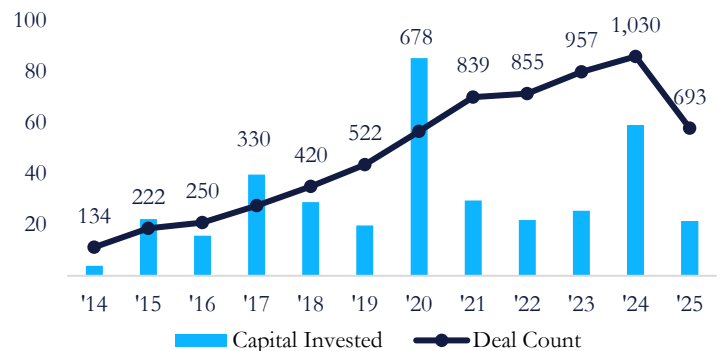
\$737 Million

KBR, Inc.

LinQuest Corporation

In September 2024, KBR, Inc. closed its acquisition of LinQuest Corporation. KBR is a Texas-based science and engineering solutions firm, providing services to U.S. and allied governments with a recent focus on defense, space, and intelligence solutions. LinQuest is a Virginia-based security contractor, with a specialization in space system engineering, digital transformation, and mission operations for the U.S. space and intelligence community at large.

Capital Invested (in \$B) and Deal Count



Historically, KBR's involvement with government customers was focused on their specialized expertise in construction, engineering, and mission solutions. In the years immediately preceding the acquisition, KBR had been awarded contracts with NASA for various solutions in space exploration. LinQuest brings to KBR deep technical capabilities in space systems engineering, along with a highly cleared workforce, providing a clear path for KBR to expand past their standard scope of work. Prior to the merger, LinQuest had contracts with the U.S. Space Force and the U.S. Air Force for \$500 million and \$562 million with purposes of analytical support and satellite development. Since becoming a KBR subsidiary, LinQuest has received a \$970 million contract with Space Systems command for digital engineering, and a \$98.7 million contract with the Space Force to provide a digital engineering ecosystem. The downstream contracts underscore KBR's growth in space systems, positioning them to secure next generation contracts.

SATELLITES IN TELECOM

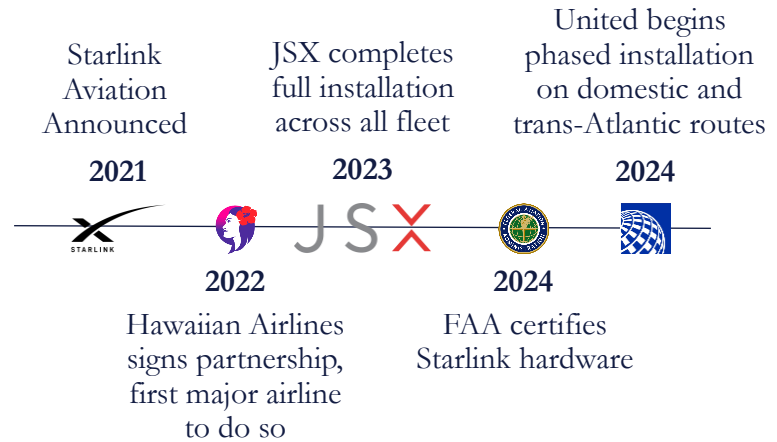
Overview

The convergence of telecommunications and space technology rapidly alters how consumers and enterprisers are connecting with one another. The introduction of low-Earth orbit (LEO) satellite constellations (as opposite to cell towers or fiber/undersea cables) allows companies to provide mobile and broadband connectivity to users anywhere possible. This leads to extended service coverage and lowered long-term infrastructure costs. Airlines, cloud companies, telecom operators, and emerging markets are all potential benefiterers of LEO.

AT&T x AST SpaceMobile

AT&T's (NYSE: T) strategic collaboration with AST SpaceMobile (NASDAQ: ASTS) is an ambitious one, aiming to build the first space-based cell broadband network. AST's BlueWalker and BlueBird satellites will act as "cell towers" and transmit cell signal over large geographic areas. AST was able to demonstrate two-way voice call and video stream in 2023, showcasing the feasibility of satellite-to cell connectivity. AT&T helps to provide customer base and the terrestrial spectrum rights, while gibing them enhanced coverage. AST gets scale and validation from this collaboration. The companies have planned to start phasing rollouts imminently, pending FCC clearance.

Timeline of Starlink x Airlines



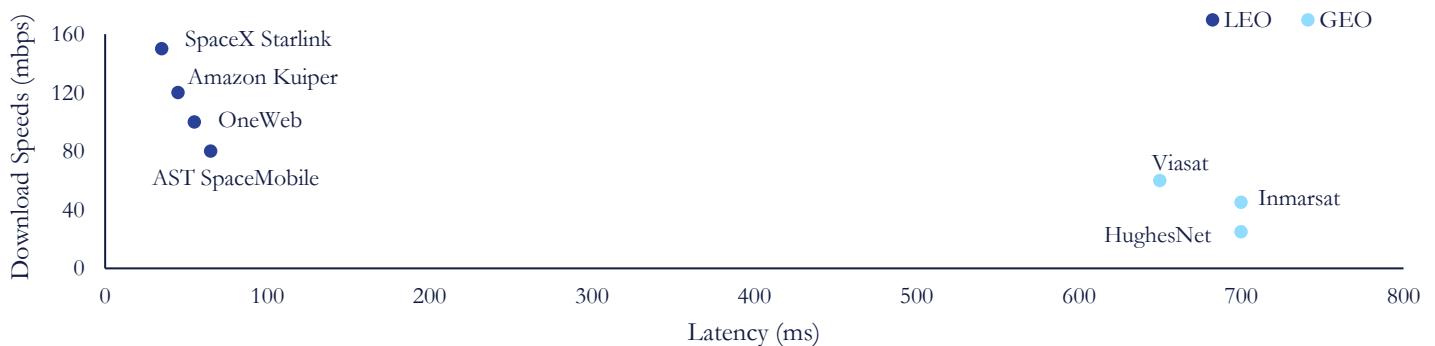
Verizon x Amazon Kuiper

Verizon (NYSE: VZ) is working with Amazon's (NASDAQ: AMZN) Project Kuiper to strengthen their 5G backhaul and enterprise broadband capabilities. Kuiper will have a constellation of 3,236 satellites, will bring high-speed and low-latency connectivity worldwide. Verizon plans to use Kuiper's LEO argument to strengthen its wireless and private 5G offerings, with a focus on rural enterprise companies and government contracts. Amazon Web Services also stands to benefit from this relationship, as it plans to integrate Kuiper into its cloud backbone for several key offerings.

Starlink x T-Mobile & Airlines

SpaceX's Starlink is expanding rapidly into consumer mobility and enterprise communications through key partnerships. Its collaboration with T-Mobile (NYSE: TMUS) integrates satellite and terrestrial networks, using Starlink's LEO constellation and T-Mobile's spectrum to enable direct satellite-to-device connectivity. Starlink is also transforming in-flight connectivity, with airlines like United, Qatar, and Hawaiian adopting its broadband systems that deliver speeds above 100 Mbps. By 2026, United and Hawaiian plan to offer free Wi-Fi powered by Starlink, cementing its role as a universal connectivity provider linking land and air networks.

Download Speed vs Latency for GEO & LEO Providers



LARGE-CAP A&D INVESTMENT

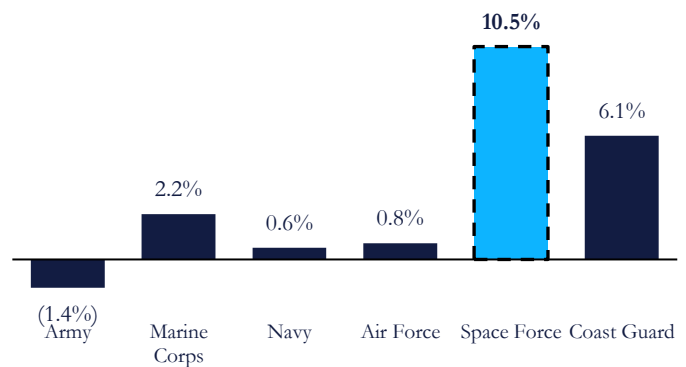
Contracts, Acquisitions, Investment

Aerospace and defense contractors have accelerated expansion past the stratosphere, making strides to evolve into full-spectrum space contractors. Fueled by growing domestic and foreign investment in the space sector, large-cap defense firms are expanding space portfolios through targeted acquisitions and strategic partnerships, positioning themselves for the next wave of government and commercial contracts. Collectively, these moves are indicative of a lasting change in defense as space becomes a core focus of both national security and commercial innovation.

Boeing (NYSE: BA)

While best known for their commercial and military aircraft offerings, Boeing has a long-standing presence in the satellite and space system sector. The firm has recently announced a \$2.8 billion dollar commitment from the U.S. government as part of the Evolved Strategic Satellite Communications program, expanding U.S. nuclear architecture. Boeing is slated to deliver 2 satellites with opportunity for 2 more in the future. Beyond this program, Boeing had reinforced its space presence in 2024 with involvement in the U.S. Space Force's Space Systems Command, securing two contracts totaling over \$1 billion. The firm has proven leadership in space with drivers for further success.

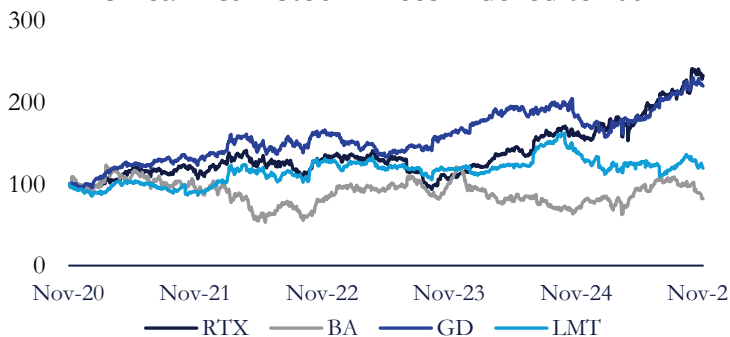
% Change In Active Personnel by
Military Agency Since 09/30/2023



RTX Corporation (NYSE: RTX)

RTX has rapidly expanded its platform in the space domain through a recent series of acquisitions. Since 2020, the firm has acquired Blue Canyon Technologies, a leading small-satellite manufacturer; SEAKR engineering, a space electronics and processing system manufacturer; and the space domain & orbital services company NORSS. These acquisitions position RTX to deliver vertically integrated satellite platforms across multiple orbits. Since then, RTX and its newfound subsidiaries have been involved in projects with NASA, DARPA, and the UK space agency, with involvement in the full space mission lifecycle.

5 Year A&D Stock Prices Indexed to 100



Lockheed Martin (NYSE: LMT)

Lockheed Martin has reinforced its established leadership in the space domain. In 2024, the firm acquired Terran Orbital for \$450 million. Prior to this acquisition, Lockheed had partnered with the satellite manufacturer on Space Development Agency programs and various Lockheed technology demonstrations. The acquisition integrates a key designer and manufacturer of modular spacecraft and streamlines Lockheed's future production capabilities

General Dynamics (NYSE: GD)

General Dynamics has historically created value through air and sea manufacturing, but the defense contractor is making strides in the space sector. The firm has differentiated itself from defense contracting competitors with ground infrastructure and network operations, securing contracts with NASA for ground communication modernization, and a \$324 million contract in conjunction with Iridium Communications for the Space Development Agency creating earth side infrastructure. While many look to reach the stars, GD has established a foothold in managing them.

RECENT NEWS

Geospatial Intelligence Funding Reaches New Heights

Investments into geospatial intelligence (GEOINT) has hit a new high of \$21.6B in Q3, driven by AI and defense demand. Governments are increasingly prioritizing real-time Earth observation, threat detection, and space domain awareness, positioning GEOINT as a mission-critical layer of international security infrastructure. As governments aim to carry out “Digital ID” systems and partner with companies such as Palantir for advanced tracking and monitoring of not only threats but of our everyday lives, GEOINT will continue to see rapid growth supplemented by AI.

Blue Origin’s New Glenn Rocket Sticks Commercial Landing on Mars

Blue Origin had a successful launch of its New Glenn rocket on its first customer mission. The vehicle carried two Mars-bound NASA satellites, making this Blue Origin’s first deep-space mission and proving themselves as viable challenger to SpaceX’s dominance in such launches. After deploying the payloads, the reusable booster successfully landed on an ocean platform, which shows the reusability and cost-saving potential for the rocket. Overall, this mission is a large turning point for the company as it positions New Glenn as a genuine option when it comes to both government and commercial clients. It also supports Blue Origin’s broader goal of building a fully reuseable space-transportation ecosystem.

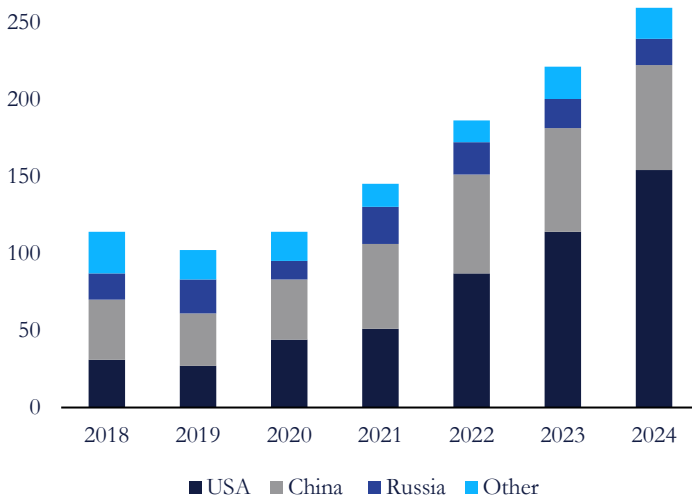
Velo3D & iRocket Bolster Relationships

Velo3D announced an expanded partnership with reusable rocket and solid rocket manufacturer iRocket to apply advanced metal additive manufacturing (AM) in next-gen aerospace/space parts. AM is central to space manufacturing (less material waste, ability to build complex geometry, integration of functions). This partnership aims to allow iRocket to have a turnaround time of under 24 hours for each launch vehicle. The simplification and streamlining of iRocket’s complex manufacturing processes serves as evidence for terrestrial manufacturing innovations enabling future in-orbit production.

Firefly’s Recent Struggles

Since their August IPO, Firefly has seen stock price fall by 75%. Stock has fallen sharply since its IPO due to repeated launch failures, rising execution risk, and concerns that delays will impair backlog conversion and cash flow visibility. After a failed Alpha Flight 7 launch, in which the rocket exploded, Firefly saw stock plummet 21% overnight, showing considerable fear from investors after repeated launch failures. This selloff reflects market sensitivity to operational failures in capital-intensive space companies, underscoring the need for successful upcoming results to rebuild investor confidence.

Orbital Launches Yearly By Country



Firefly Price Action



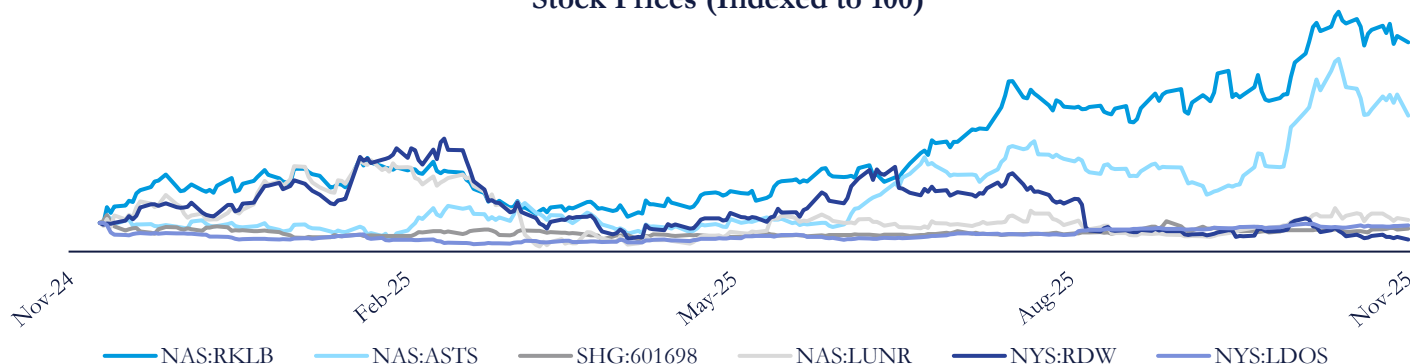


BULL & BEAR
RESEARCH

Financial Analysis

Logo	Company	Ticker	Description	Price (USD)
	Rocket Lab	RKLB	Aerospace company offering end-to-end space systems, such as launch services, manufacturing of spacecrafts, and satellite components. Operates the <i>Electron</i> small-satellite launch vehicle and is currently developing <i>Neutron</i> rocket, which is larger and partially reusable. In addition, is currently building out a suite of spacecraft buses, satellite subsystems,, and mission operations capabilities.	\$40.30
	AST SpaceMobile	ASTS	Satellite communications company focused on building space-based cellular broadband network, which would let mobile phones directly connect to satellites. <i>SpaceMobile</i> , its constellation and <i>BlueWalker/BlueBird</i> , its prototypes are designed to fill global connectivity gaps, notably in remote/underserved regions.	\$51.37
	China Satellite	601698	State-owned Chinese satellite operator under CASC (China Aerospace Science and Technology Corporation). Provides satellite communications and broadcasting services both across and China and beyond, offering bandwidth leasing, infrastructure for broadcast/telecom clients, satellite resource services, and telemetry/ground station support.	\$20.18
	Intuitive Machines	LUNR	Space exploration and infrastructure company with a focus on lunar services. Offerings include payload delivery to the moon, enablement of infrastructure for sustained lunar operations, as well as communications/data relay networks. Holder of several NASA contracts under Commercial Lunar Payload Services program.	\$8.62
	Redwire Corp.	RDW	Space technology firm that offers mission support for civil, commercial, and defense customers, as well as produces high-reliability components and spacecraft systems. Offerings include satellite payloads, in-space manufacturing/assembly, deployable structures, and sensors, all with overall aim to be “mission partner of choice” in space economy.	\$5.30
	Leidos Holdings	LDOS	Major U.S. tech/defense contractor that provides systems integration, engineering, IT, and analytics services to national security and intelligence agencies, as well as civil. Supports space tech and programs vis digital and systems architecture capabilities.	\$186.49

Stock Prices (Indexed to 100)



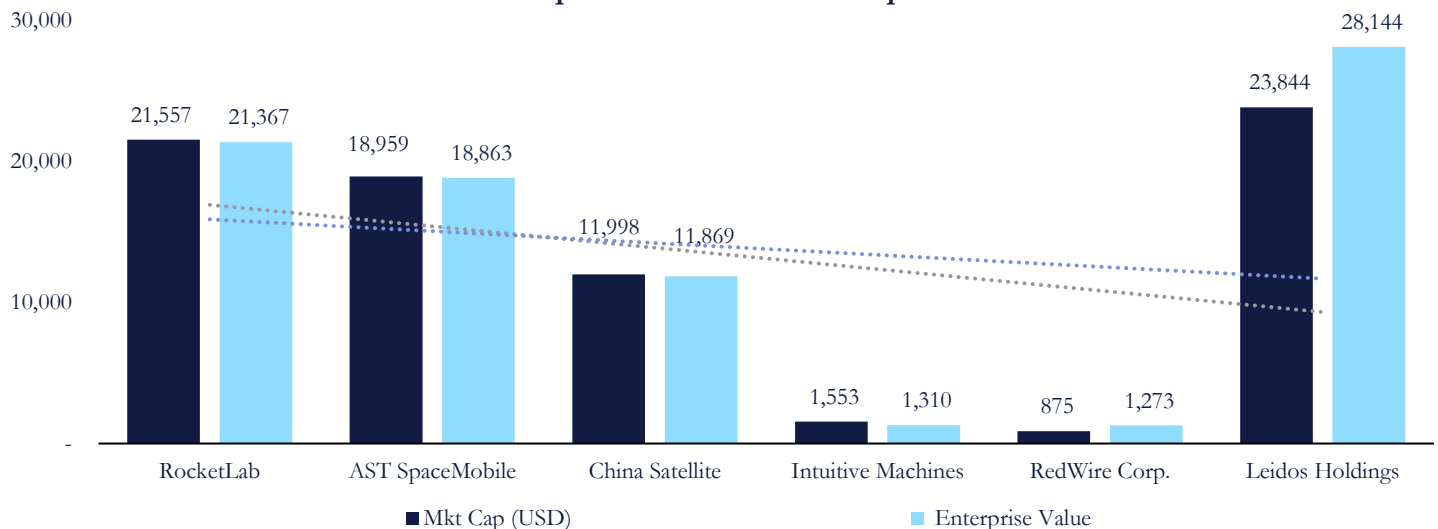
PEER UNIVERSE OVERVIEW

\$ in millions of USD (As of Q3 2025)

Company	Mkt. Cap	Enterprise Value	Share Price	TTM		TEV /		EPS	P/E
				Revenue	EBITDA	Revenue	EBITDA		
RocketLab	21,557	21,367	40.30	555	(179)	34x	--	(0.36)	--
AST SpaceMobile	18,959	18,863	51.37	19	(229)	952x	--	(1.24)	--
China Satellite	11,998	11,869	20.18	365	--	34x	--	0.01	329x
Intuitive Machines	1,553	1,310	8.62	220	(65)	7x	--	(2.40)	--
Redwire Corp.	875	1,273	5.30	296	(101)	6x	--	(2.87)	--
Leidos Holdings	23,844	28,144	186.49	17,332	2,372	2x	12x	10.95	17x
Lower	875	1,273	5.30	19	(229)	2x	12x	(2.87)	17x
Mean	13,131	13,804	52.04	3,131	360	172x	12x	0.68	173x
Median	15,478	15,366	30.24	331	(101)	21x	12x	(0.80)	173x
Upper	23,844	28,144	186.49	17,332	2,372	952x	12x	10.95	329x

The Peer Universe Overview table provides a high-level snapshot of key financial metrics across companies in the space-manufacturing technology sector as of 11/21/25. Among this group, Leidos Holdings stands out with the largest market cap, enterprise value, and revenue base, and is the only peer generating positive EBITDA. We evaluate these companies similarly to high-growth technology firms: not on near-term profitability, but on revenue scale and long-term growth runway. In this context, EV/Sales becomes the most relevant metric, as it reflects what the market is willing to pay for each dollar of current revenue relative to enterprise value. AST SpaceMobile, despite limited revenue today, trades at an outsized 952x EV/Revenue, signaling that investors are betting heavily on future commercialization and multiple normalization. China Satellite and Redwire, while less extreme, also trade at an elevated 34x EV/Revenue, reflecting constructive sentiment and willingness to hold through early-stage revenue development. Importantly, most peers remain unprofitable and therefore lack meaningful P/E ratios. While an uninformed view might suggest these companies are overvalued, we believe the combination of strong macro tailwinds, expanding space-economy demand, and substantial long-term revenue potential supports current valuations.

Market Capitalization versus Enterprise Value



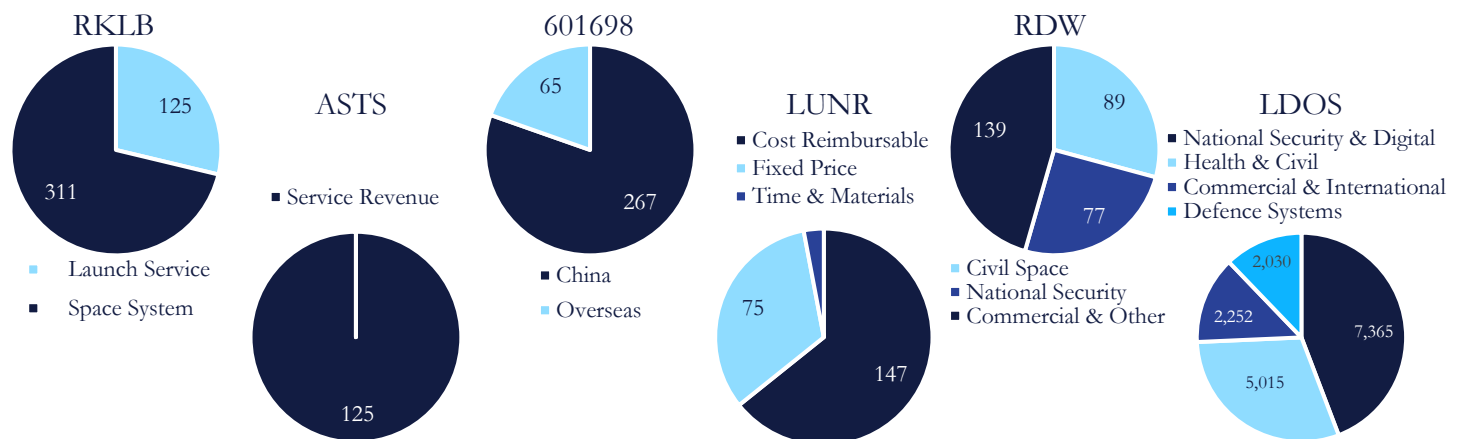
COMPARABLE: BASIC

\$ in millions of USD (As of Q3 2025)

Company	Value of Government Contracts Awarded	Successful Launches	Public vs Private Exposure	R&D Spend	Mission Success Rate
RocketLab	1,800	72	50/50	122.3	94.7%
AST SpaceMobile	43	6	70/30	28.8	100%
China Satellite	--	500+	100/0	11.3	~96%
Intuitive Machines	117	1	80/20	--	50%
Redwire Corp.	45	--	60/40	6.1	--
Leidos Holdings	738	--	87/13	--	--
Lower	43	1	50/50	6.1	50%
Mean	548.6	144.8	74.5/25.5	20.1	85.2%
Median	117	39	70/30	20.1	95.4%
Upper	1,800	500+	100/0	122.3	100%

The space technology peer set displays extreme dispersion in operational maturity, funding sources, risk tolerance, and execution profiles. Rocket Lab dominates the public pure-play segment with 72 successful orbital launches and a 94.7% lifetime success rate, while securing approximately \$1.8 billion in recent U.S. government contract awards (anchored by the critical NSSL Phase 3 Lane 1 award) and maintaining a balanced 50/50 government-to-commercial revenue mix that underscores its unique dual-role as both a high-cadence commercial provider and a trusted national security asset. By contrast, AST SpaceMobile (6 launches, 100% success) and Intuitive Machines (1 successful mission, 50% overall) remain heavily government-dependent at 70/30 and 80/20 exposure, respectively. At the opposite extreme, China's state-owned launch monopoly (CASC) has completed over 500 successful missions with ~96% reliability, operating almost entirely on state funding (100/0). Redwire and Leidos, both launch-agnostic, focus on components, integration, and defense services, exhibiting strong government reliance (60/40 and 87/13) and effectively near-zero pure R&D intensity. R&D spend ranges from \$122 million at Rocket Lab (driven by Neutron development) to \$6–29 million at smaller peers and near zero at incumbents. This stark divergence reflects a broader industry shift: capital-intensive U.S. new-space firms are burning cash to capture massive commercial and strategic markets, while state-backed and legacy services giants prioritize stability and near-term cash generation.

Revenue Mix

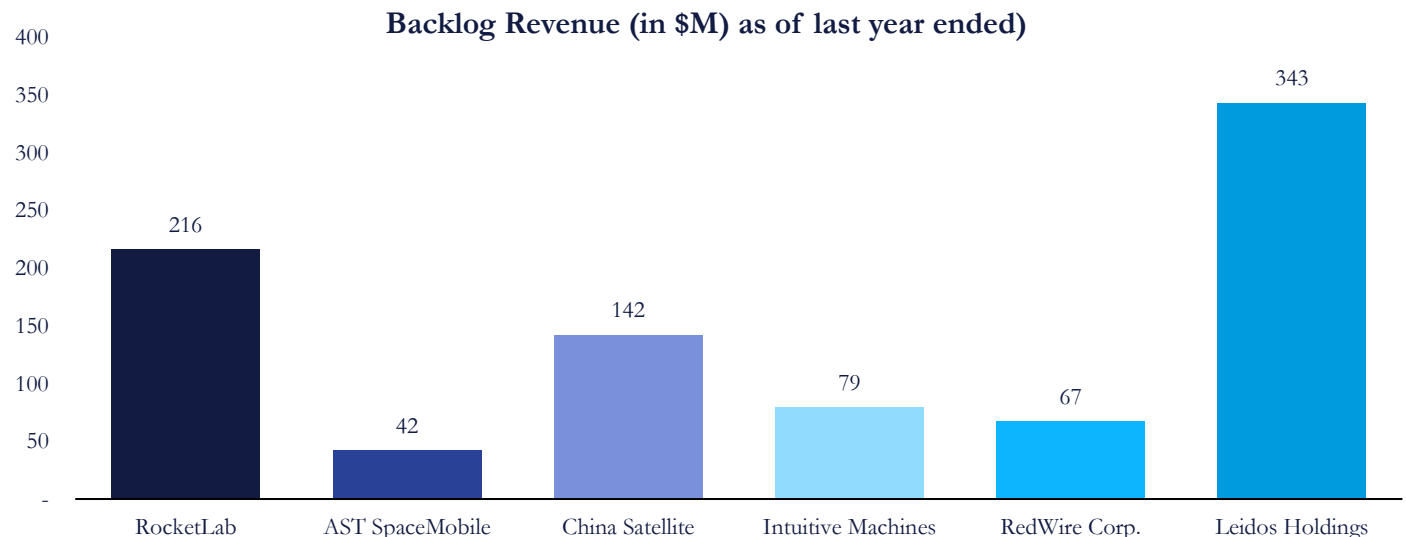


COMPARABLE: VALUATION

\$ in millions of USD (As of Q3 2025)

Company	Price/ Book	EV/ Revenue	Backlog (\$)	EV/ CapEx	Revenue Growth (YoY %)	Gross Margin	Forward Revenue Growth
RocketLab	28x	34x	216	229x	48.0%	26.6%	53.1%
AST SpaceMobile	15x	952x	42	25x	1239.9%	--	1512.2%
China Satellite	5x	34x	142*	52x	3.5%	27.7%	--
Intuitive Machines	--	7x	79	42x	(10.3%)	1.2%	147.5%
Redwire Corp.	1x	6x	67	93x	50.7%	14.6%	56.9%
Leidos Holdings	5x	2x	343	168x	6.7%	16.8%	3.0%
Lower	1x	2x	42	25x	(10.3%)	1.2%	3.0%
Mean	11x	172x	148	102x	223.1%	17.4%	354.6%
Median	5x	21x	111	73x	27.3%	16.8%	56.9%
Upper	28x	952x	343	229x	1239.9%	27.7%	1512.2%

Backlog revenue is one of the most critical indicators for companies in this sector. Unlike consumer staples, these firms operate on long production cycles, often more complex and drawn out than commercial aerospace, so revenue materializes over years, not quarters. In this context, Leidos' substantial backlog stands out, especially given its relatively modest valuation at ~2x EV/Revenue. Capital expenditures also offer a valuable window into future growth. Lower EV/CapEx multiples generally imply that a company is reinvesting heavily relative to its enterprise value. AST SpaceMobile is a clear example: despite limited investor conviction today, its ~25x EV/CapEx multiple reflects meaningful investment into scaling operational capacity. By contrast, Rocket Lab trades at ~229x EV/CapEx, a potential red flag when paired with its lower R&D intensity. Gross margin performance, where Rocket Lab and China Satellite currently lead, is less informative at this stage, as many of these companies are pre-scale, pre-catalyst, and valued primarily on forward revenue potential rather than current profitability metrics. Overall, we believe these companies are priced on long-duration growth expectations. AST SpaceMobile, Intuitive Machines, Redwire, and Rocket Lab all reflect the market's view that early leaders in commercial space have significant upside as their backlogs convert, capacity expands, and the broader space economy matures.



Sources: Bloomberg, Company Filings, Morningstar

* = China Satellite does not report quarterly; therefore, the backlog represents the 2024 end-of-year value

COMPARABLE: PROFITABILITY

\$ in millions of USD (As of Q3 2025)

Company	Net Income Margin	Revenue / Employee	Operating Margin	Asset Turnover Ratio	CapEx Intensity (x)	R&D Intensity (x)
RocketLab	(11.8%)	264,064	(41.3%)	0.4x	2.8x	0.4x
AST SpaceMobile	(833.7%)	32,061	(540.6%)	0.0x	137.7x	1.3x
China Satellite	16.7%	617,037	18.6%	0.1x	8.9x	0.0x
Intuitive Machines	(13.0%)	505,600	(29.4%)	0.4x	3.4x	0.0x
Redwire Corp.	(39.8%)	394,863	(40.5%)	0.3x	4.9x	0.0x
Leidos Holdings	8.2%	361,083	12.0%	1.3x	0.8x	0.0x
Lower	(833.7%)	32,061	(540.6%)	0.0x	0.8x	0.0x
Mean	(145.6%)	362,451	(103.5%)	0.4x	26.4x	0.3x
Median	(12.4%)	377,973	(34.9%)	0.4x	4.2x	0.0x
Upper	16.7%	617,037	18.6%	1.3x	137.7x	1.3x

The selected space technology companies exhibit sharply divergent profitability and operational efficiency, with Net Income Margins ranging from +16.7% to (833.7%). China Satellite stands alone as the only consistently profitable peer, supported by stable government-related revenue and an 18.6% operating margin, whereas AST SpaceMobile (833.7%), Redwire (39.8%), and Intuitive Machines (13.0%) remain firmly in investment mode, prioritizing constellation build-out, launch infrastructure, and contract execution over near-term earnings. Revenue per Employee ranges widely from \$617k at China Satellite to \$324k–\$505k among the U.S. growth names, with AST SpaceMobile delivering respectable productivity despite its heavy burn. Asset turnover is uniformly low (median 0.4x), typical of a sector where large upfront investments precede revenue recognition by several years. R&D intensity remains subdued as most peers have either moved beyond peak research or embed development within project-specific CapEx. Ultimately, the wide dispersion across margins, productivity, and investment levels underscores the classic space technology trade-off: current negative profitability and sustained cash burn in exchange for strategic positioning in disruptive, multi-billion-dollar addressable markets spanning satellite broadband, low-cost launch, and in-space servicing, with China Satellite representing the mature cash-generative model and the U.S. peers embodying capital-intensive high-growth profiles still years from sustained profitability.

Supplier & Customer Mix

RocketLab <i>Top Suppliers</i> 	China Satellite <i>Top Suppliers</i> 	RedWire Corp. <i>Top Customers</i> 
AST SpaceMobile <i>Top Suppliers</i> 	Intuitive Machines <i>Top Suppliers</i> 	Leidos Holdings <i>Top Suppliers</i> 

BBR

BULL & BEAR
RESEARCH

Initiating Coverage: Intuitive Machines



Intuitive Machines (LUNR)

Intuitive Machines is an American aerospace company founded in 2013. Their mission is to commercialize the Moon: Delivering payloads, providing lunar infrastructure, data transmission and logistics services. LUNR develop landers that carry scientific and commercial payloads to the lunar surface.

Differentiation and Strategy

- First U.S. private lunar lander:** Early Nova-C platform giving surface access.
- End-to-end capability:** Unlike most peers, they own the full stack: design, manufacturing, mission operations, and lunar communications.
- Transition to recurring revenue:** Shifting from one-off lander missions to continuous “Lunar Infrastructure-as-a-Service” (Data, navigation, and power)
- Dual exposure:** Serves both NASA’s Artemis program and commercial clients, positioning it between public funding and private market growth.
- LUNR is strategy is composed of 3 main drivers: Leverage NASA contracts to prove capability and fund early missions. Expand vertically into communication satellites and lunar infrastructure to control costs and margins. Finally, diversifying horizontally into cislunar and EO services building diverse income.

Financial Performance

LUNR has shown rapid top-line growth but struggles with operating leverage. TTM revenue is roughly \$226 million, a sharp increase YoY driven by NASA CLPS contracts and commercial payload activity. In terms of profitability, the company remains in a tough spot. LUNR is still in a building phase, so BBR does not view the negative profit as a bad thing and have modeled it out in comparison more so as a combination of a high growth tech company, and an A&D company. Backlog remains healthy following several new government and commercial awards, providing near-term revenue visibility.

Recent Acquisitions

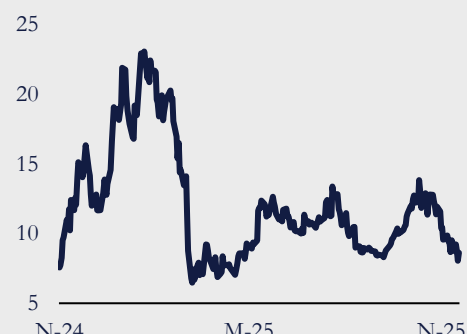
- Lanteris Space Systems:** In November 2025, LUNR announced they will acquire Lanteris from PE firm Advent Intl. for \$800 million. Lanteris brings large-scale spacecraft manufacturing capabilities (satellite buses, missions in orbit), expanding LUNR footprint beyond lunar surface delivery.
- KinetX:** Acquired in August for around \$30 million. KinetX is known for deep-space navigation and will enhance LUNR’s executing complex missions.

Outlook

We have a **buy** rating on LUNR, with a price target of \$11.92, a 24% upside from its current valuation. The company's management has recognized their lack of revenue diversification, and the stock is undervalued due to its current operating expenses. As the space business continues to expand, we see LUNR emerging as a clear winner, and provider of services across a multitude of verticals in Space Manufacturing, and Mission execution.

Stock Rating	Buy
BBR Price Target	\$11.92
Price (11/21/25)	\$8.62
Ticker	LUNR
Exchange	NYSE

52 Week Range \$6.13 – \$24.95



Research Team

Miles Rucker
Owen Romano
Zachary Cohen
Rohan Kalaria
Gonçalo Soares

	Multiples	
	LUNR	Peers
Sales Growth	7.97%	11.38%
Debt/Assets	10.5%	20.5%
EV/EBITDA +2	56.4x	15.1x

Stock Information

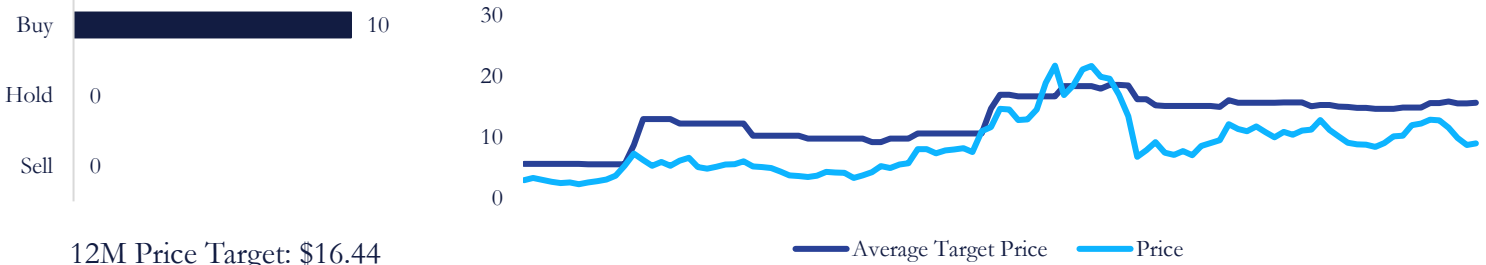
Market Cap (B)	\$1.55B
Shares Outs. (mm)	1.99
Free Float (%)	29.6%
Beta	1.42

ALTERNATIVE VALUATIONS

Consensus Rating



Price Analysis (3Y, USD)



Firm	Analyst	Date	Analyst Rating	Price Target
 B RILEY Securities	Michael Crawford	11/17/25	Buy	\$14.00
 ROTH	Suji Desilva	11/14/25	Buy	\$16.00
 CANTOR Fitzgerald	Andres Sheppard	11/14/25	Overweight	\$16.00
 cg/ Caracore Geniality	Austin Moeller	11/6/25	Buy	\$15.50
 Deutsche Bank	Edison Yu	11/5/25	Buy	\$18.00
 CH Craig Hallum	Jonathan Siegmann	11/5/25	Buy	\$18.00
STIFEL	Jeff Van Rhee	11/5/25	Buy	\$17.00
 Clear Street	Edison Yu	11/4/25	Buy	\$17.00
 Baptista Research	Ishan Majumdar	10/4/25	Buy	\$15.80

Select Commentary

Deutsche Bank

Price Target: \$18.00

Rating: Buy

- Acquisition of Lanteris provides reason for optimism, Lanteris generating nearly 3x more revenue on a TTM basis and EBITDA margin improved from single digits to double digits, suggesting a much healthier underlying business now
- Lower 2025E forecasts, now calling for revenue of \$218m (-4% YoY), gross margin of 2.4%, and EBITDA loss of \$50m. Looking at 2026E, embed 3 quarters of Lanteris contribution, leading to revenue/EBITDA of +\$830m/\$35m
- For valuation, pivot to a blended multiple approach using 5x 2030E EV/revenue and 15x 2030E EV/EBITDA (in line with more established A&D companies). Reiterate Buy rating and \$18 price target.

Research Team

Edison Yu

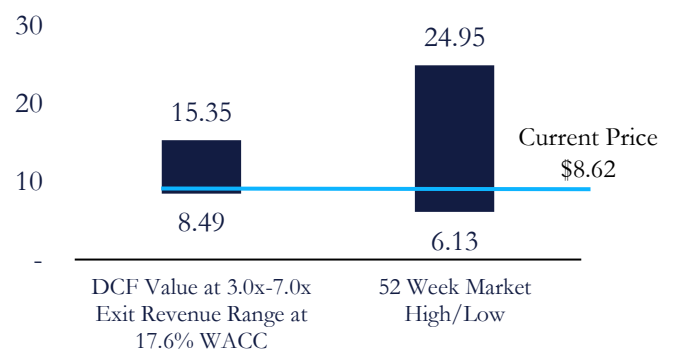
Laura Li



VALUATION SUMMARY

Income Statement						
Fiscal Year	2025 E	2026 E	2027 E	2028 E	2029 E	2030 E
Year Ended	12/31/2025	12/31/2026	12/31/2027	12/31/2028	12/31/2029	12/31/2030
Revenue	262	302	347	381	420	462
Cost Of Goods Sold	220	222	221	243	268	294
Gross Profit	43	79	126	138	152	167
Selling General & Admin Exp.	53	52	50	43	35	39
Depreciation & Amort.	13	16	19	22	25	28
Other Operating Exp., Total	66	69	69	65	60	67
Operating Income	(24)	10	57	73	92	101
EBT Incl. Unusual Items	(24)	10	57	73	92	101
Income Tax Expense	4	2	9	11	14	15
Net Income	(28)	9	48	62	78	85
Balance Sheet						
Fiscal Year	2025 E	2026 E	2027 E	2028 E	2029 E	2030 E
Cash And Equivalents	164	170	222	294	384	486
Accounts Receivable	88	97	107	112	118	123
Prepaid Exp.	5	6	6	7	8	8
Net Property, Plant & Equipment	89	105	111	118	124	130
Total Assets	349	379	448	533	635	750
Accounts Payable	20	23	26	32	39	47
Accrued Exp.	17	18	18	17	15	17
Unearned Revenue, Current	70	74	78	79	78	77
Long-Term Leases	35	35	35	35	35	35
Unearned Revenue, Non-Current	14	13	11	13	14	15
Other Non-Current Liabilities	203	203	203	203	203	203
Total Liabilities	359	366	372	378	384	394
Total Equity	(10)	13	76	155	251	355
Total Liabilities And Equity	349	379	448	533	635	750
Discounted Cash Flow Valuation						
Fiscal Year	2025E	2026E	2027E	2028E	2029E	2030E
Revenue	262	302	347	381	420	462
EBIT	(24)	10	57	73	92	101
NOPAT	(20)	9	48	62	78	85
D&A	13	16	19	22	25	28
Capex	(40)	(31)	(26)	(28)	(31)	(34)
Change in NWC	1	(2)	(4)	(1)	(1)	2
Unlevered FCF	(47)	(8)	38	55	71	82
Exit Multiple Approach						
Terminal year Revenue	462					
Revenue multiple	5.0x					
Terminal value in 2030	2,308					
Present value of terminal value	1,013					
Present value of stage 1 cash flows	87					
Enterprise value (stage 1 + 2)	1,100					
Enterprise value	1,100					
Net debt	(308)					
Equity value	1,408					
Shares outstanding	118					
Equity value per share	\$11.96					

Equity Valuation using DCF





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RESEARCH

Initiating Coverage: Rocket Lab

Rocket Lab USA, Inc. (RKLB)

Rocket Lab is a leading end-to-end space company providing launch services, spacecraft manufacturing, and on-orbit operations. The company operates the Electron Rocket, the second most launched U.S. rocket after SpaceX's Falcon9. Beyond launch operations, Rocket Lab offers satellite buses, components, and mission operations.

Differentiation and Strategy

Integrated operational advantage: Vertical integration across launch, spacecraft, software, and mission operations gives Rocket Lab full control of the mission lifecycle, faster execution, and lower cost than competitors.

Shift toward recurring revenue: RKLB is evolving beyond launch services into higher-margin space systems, satellite components, and in-orbit mission services, positioning themselves for durable revenue streams.

Strategic dual exposure: Positioned to serve both commercial constellations and defense clients — benefiting from rising national security spending while retaining access to commercial satellite and payload missions.

Financial Performance

Rocket Lab reported approximately \$550 million in TTM revenue, reflecting ~25% year-over-year growth supported by accelerating demand in Launch Services and higher-margin Space Systems sales. Profitability remains challenged as the company continues to invest in R&D and infrastructure, but management has guided toward significant revenue growth in 2025. Management is targeting positive free cash flow and net income for the first time by FY 2026, supported by reduced cash burn, improved cost discipline, and growing exposure to scalable mission services.

Recent Acquisitions

GEOST (March 2025): Acquired for \$275 million, GEOST provides space-based optical payloads and space domain awareness systems that deepen Rocket Lab's defense exposure and support the shift toward recurring mission services.

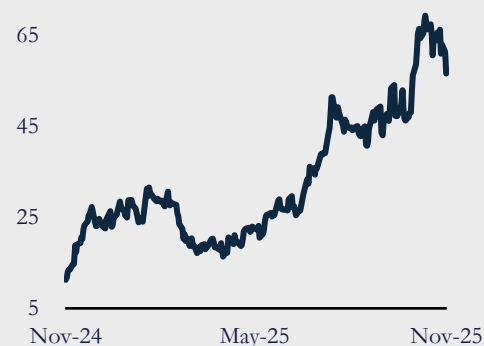
SolAero Technologies (January 2022): Acquired for ~\$80 million, SolAero enhances Rocket Lab's in-house power systems, supports long-duration missions, and boosts margins through vertical integration.

Outlook

We have a **buy** rating for RKLB, with a price target of \$50.60, a 22.46% upside from its current valuation. Rocket Lab is positioned to benefit from rising demand for defense-oriented space capabilities and commercial satellite constellations. The company is prioritizing scalable, recurring revenue through mission services and satellite components, which carry higher margins than launch operations. A major catalyst lies in the development of the Neutron rocket: a reusable medium-lift launch vehicle expected to compete directly with SpaceX and ULA.

Stock Rating	Buy
BBR Price Target	\$50.60
Price (11/24/25)	\$41.32
Ticker	RKLB
Exchange	NSDQ

52 Week Range \$14.72 – \$73.97



Research Team

Miles Rucker
Owen Romano
Zachary Cohen
Rohan Kalaria
Gonalo Soares

Multiples

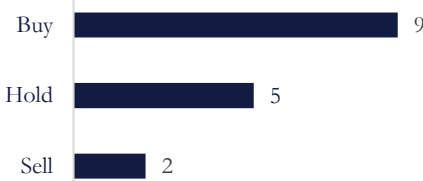
	RKLB	Peers
Sales Growth	48%	27%
EV/ Rev	34x	21x
Cash Burn	40x	28x

Stock Information

Market Cap (B)	\$21.53
Shares Outs. (mm)	479.36
Free Float	95.4%
Beta	2.17

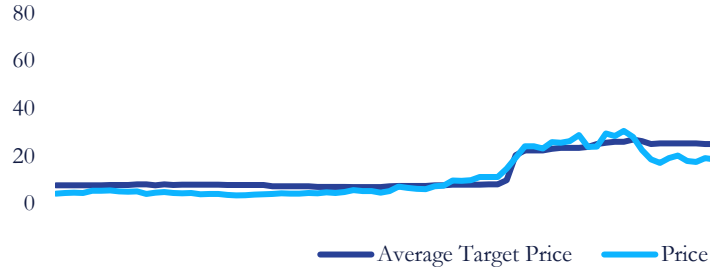
ALTERNATIVE VALUATIONS

Consensus Rating



12M Price Target: \$64.69

Price Analysis (3Y, USD)



Firm	Analyst	Date	Analyst Rating	Price Target
 Goldman Sachs	Anthony Valentini	11/13/25	Neutral	\$47.00
 TD	Gautam Khanna	11/13/25	Buy	\$60.00
Morgan Stanley	Kristine Tan Liwag	11/12/25	Equalweight	\$47.00
 Needham	Ryan Koontz	11/12/25	Buy	\$63.00
 BAIRD	Peter J Arment	11/12/25	Outperform	\$83.00
 Citizens	Trevor Walsh	11/12/25	Market Perform	--
 Deutsche Bank	Edison Yu	11/11/25	Buy	\$55.00
 KeyBank	Michael Leshock	11/11/25	Overweight	\$75.00
 CANTOR Fitzgerald	Andres Sheppard	11/11/25	Overweight	\$72.00

Select Commentary

Needham & Company

Price Target: \$53.00

Rating: Buy

- Come away more bullish on Electron/Haste opportunities, evidenced by strong new Int'l bookings, and opportunities to win large gov't satellite contracts, including SDA Tranche 3. We increase our F25 revenue est and maintain F26 flat, while reducing F25/F26 EBITDA on higher opex.
- RKL B expects 4Q25 revenue of \$170MM-\$180MM (MP above consensus \$172MM) and NG GM to be in the range of 43%-45%, above the 38.5% consensus. RKL B expects 4Q25 Adj. EBITDA of (\$29MM)-(\$23MM) and NG Opex between \$107MM-\$113MM.
- We also raise our PT to \$63 (from \$55) on strong execution and promising multi-year opportunities in both Space Systems and Launch.

Research Team

Ryan Koontz

Matt Cavanagh

Jeffrey Hopson



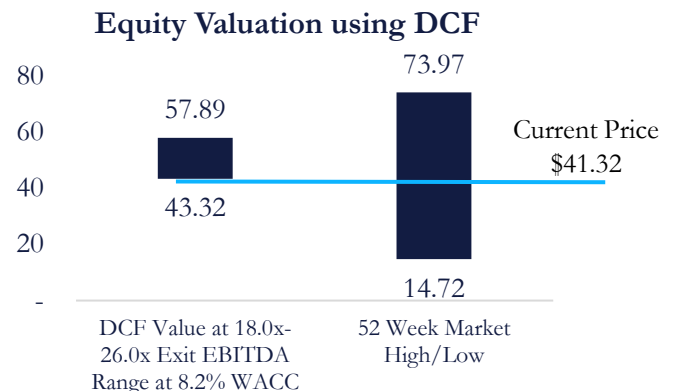
VALUATION SUMMARY

Income Statement						
Fiscal Year	2025 E	2026 E	2027 E	2028 E	2029 E	2030 E
Year Ended	12/31/2025	12/31/2026	12/31/2027	12/31/2028	12/31/2029	12/31/2030
Revenue	580	870	1,218	1,583	2,027	2,432
Cost Of Goods Sold	339	421	467	528	575	568
Gross Profit	241	449	751	1,055	1,452	1,864
Selling General & Admin Exp.	146	193	233	256	267	247
R & D Exp.	174	217	268	301	324	316
Operating Income	(78)	39	250	499	861	1,301
Interest Expense	(4)	(4)	(4)	(4)	(4)	(4)
EBT Incl. Unusual Items	(82)	35	246	495	857	1,297
Income Tax Expense	(12)	5	37	74	129	195
Net Income	(70)	30	209	420	729	1,102

Balance Sheet						
Fiscal Year	2025 E	2026 E	2027 E	2028 E	2029 E	2030 E
Total Cash & ST Investments	378	456	701	1,100	1,770	2,766
Accounts Receivable	130	184	240	312	400	480
Inventory	117	134	136	139	136	119
Prepaid Exp.	51	76	106	138	177	212
Total Current Assets	693	875	1,219	1,735	2,542	3,648
Net Property, Plant & Equipment	307	378	479	609	774	963
Total Assets	1,229	1,482	1,927	2,573	3,544	4,840
Accounts Payable	61	81	97	109	119	118
Accrued Exp.	42	54	66	73	78	74
Unearned Revenue, Current	276	396	530	658	802	913
Total Current Liabilities	394	555	726	882	1,052	1,170
Long-Term Debt	389	389	389	389	389	389
Total Liabilities	857	1,017	1,188	1,345	1,515	1,632
Total Equity	372	465	739	1,229	2,030	3,208
Total Liabilities And Equity	1,229	1,482	1,927	2,573	3,544	4,840

Discounted Cash Flow Valuation						
Fiscal Year	2025E	2026E	2027E	2028E	2029E	2030E
Revenue	580	870	1,218	1,583	2,027	2,432
EBIT	(78)	39	250	499	861	1,301
NOPAT	(67)	33	212	424	732	1,106
D&A	28	37	49	66	87	112
Capex	(72)	(108)	(151)	(196)	(251)	(301)
Change in NWC	30	74	73	43	30	(10)
Unlevered FCF	(81)	36	184	336	598	907

Exit Multiple Approach	
Terminal year EBITDA	1,413
EBITDA multiple	22.0x
Terminal value in 2029	31,079
Present value of terminal value	19,248
Present value of stage 1 cash flows	4,805
Enterprise value (stage 1 + 2)	24,053
Enterprise value	24,053
Net debt	(251)
Equity value	24,304
Shares outstanding	479
Equity value per share	\$50.70



A satellite with solar panels is shown in space, with a view of Earth's surface (oceans and landmasses) in the background. The satellite is white with blue solar panels and a red logo. A large white diagonal line cuts across the image from the bottom right towards the top right.

BBR

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RESEARCH

Initiating Coverage: Redwire Corporation

Redwire Corporation (RDW)

Redwire Corporation, founded in 2020, is a space and defense technology firm creating a more sustainable space economy through its integrated infrastructure solutions. It provides solar arrays, avionics, RF platforms, and in-space manufacturing to support civil, commercial, and national security missions.

Differentiation and Strategy

Full-stack multi-domain provider: Rare end-to-end owner of critical subsystems from ROSA solar arrays and deployable structures to avionics, RF systems, robotic arms, in-space manufacturing capabilities, and biotech payloads.

Integrated mission execution: Delivers complete spacecraft buses, on-orbit servicing, and life-extension solutions, reducing customer integration risk versus component-only peers.

Recurring revenue transition: Shifting from hardware sales to high-margin services, satellite refueling/repair, and on-orbit manufacturing/assembly.

Dual civil-defense exposure: Strong NASA/ESA heritage combined with Edge Autonomy’s combat-proven UAS and DARPA VLEO contracts creates balanced growth across government and national security markets.

Financial Performance

Redwire posted Q3 2025 revenue of \$103.4 million, up 50.7% YoY, driven by space infrastructure and Edge Autonomy despite government delays. Adjusted gross margin hit 27.1% and adjusted EBITDA was \$(2.6) million as the company invests in in-space manufacturing, on-orbit servicing, and multi-domain defense capabilities. These investments position Redwire at the forefront of growing commercial and national security space markets. Record backlog over \$350 million and 1.25x book-to-bill confirm strong demand for its integrated product suite and near-term visibility despite elevated leverage.

Recent Acquisitions

Edge Autonomy (June 2025): Acquired for \$925 million, this leading UAS manufacturer brings combat-proven drones, adds over \$200 million in annual revenue, shifts revenue mix toward defense, and enables integrated air-space solutions for VLEO and orbital debris removal missions.

Hera Systems (Sept. 2024): Acquired for undisclosed terms, the spacecraft developer enables RDW to support national security space missions in GEO.

Outlook

We have a **buy** rating for RDW, with a price target of \$14.55, a 170% upside from its current valuation. Management’s execution on Edge Autonomy’s integration and backlog levels offset any concerns of government funding risks. Redwire is deeply undervalued as investors underprice their hardware-centric model in the larger sector that routinely pays software-like multiples, and as integrated space technologies scale, Redwire positions itself as the leading provider across civil, commercial, and national security verticals.

Stock Rating	Buy
BBR Price Target	\$14.55
Price (11/24/25)	\$5.39
Ticker	RDW
Exchange	NYSE

52 Week Range \$5.72 – \$26.66



Research Team

Miles Rucker
Owen Romano
Zachary Cohen
Rohan Kalaria
Gonalo Soares

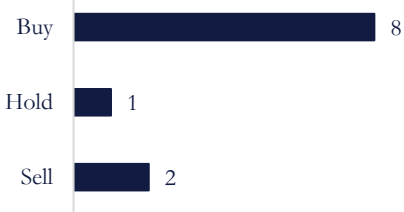
	Multiples	
	RDW	Peers
Sales Growth	50.7%	35.0%
EV/EBITDA	(3.12x)	11.4x
Price/Sales	1.7x	18.7x

Stock Information

Market Cap (M)	\$875.30
Shares Outs. (mm)	165.15
Free Float	14.87%
Beta	2.46

ALTERNATIVE VALUATIONS

Consensus Rating



Price Analysis (3Y, USD)



Firm	Analyst	Date	Analyst Rating	Price Target
	Ronald Epstein	11/19/25	Sell	\$6.00
	Austin Moeller	11/11/25	Buy	\$11.00
Jefferies	Greg Konrad	11/7/25	Buy	\$13.00
	Scott Buck	11/7/25	Buy	\$22.00
	Michael Crawford	11/7/25	Buy	\$15.00
	Suji Desilva	11/6/25	Buy	\$15.00
TRUIST 	Michael Ciarmoli	11/6/25	Hold	\$8.00
	Brian Kinstlinger	11/6/25	Buy	\$9.00
	Colin Canfield	11/5/25	Overweight	\$20.00

Select Commentary

Truist Securities

Price Target: \$8.00 Rating: **Hold**

- Maintain HOLD rating but trim our PT to \$8 from \$11 following 3Q25.
- Management lowered 2025 revenue guide at the midpoint to \$330M vs prior \$415M, 18% below the Street, as government contracting delays and award timing weighed on performance.

Research Team
Michael Ciarmoli

TRUIST 

Jefferies

Price Target: \$13.00 Rating: **Buy**

- Sell-off driven by delays in the topline with a ~20% 2025 revenue guide cut
- A positive: operating performance is steadying ahead of topline inflection, a source of potential upside as \$10BB pipeline (\$3BB of YTD submits) starts to convert

Research Team
Greg Konrad

Jefferies

VALUATION SUMMARY

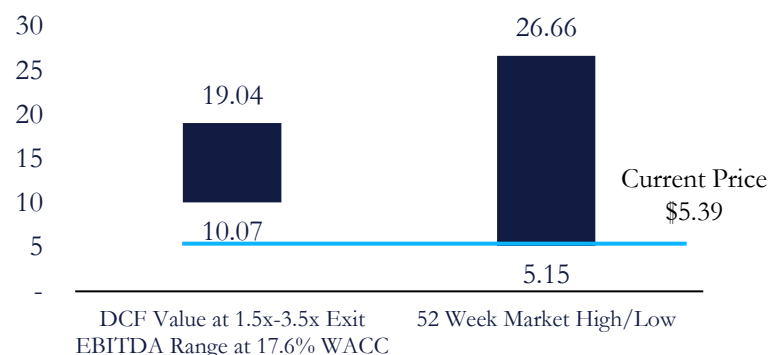
Income Statement						
Fiscal Year	2025 E	2026 E	2027 E	2028 E	2029 E	2030 E
Year Ended	12/31/2025	12/31/2026	12/31/2027	12/31/2028	12/31/2029	12/31/2030
Revenue	346	528	682	878	1,115	1,441
Cost Of Goods Sold	244	319	344	442	562	726
Gross Profit	103	209	339	435	553	715
Selling General & Admin Exp.	71	92	99	101	128	166
R & D Exp.	7	11	14	18	22	29
Operating Income	25	106	226	317	403	520
Interest Expense	(14)	(14)	(14)	(14)	(14)	(14)
EBT Incl. Unusual Items	11	93	213	303	389	507
Income Tax Expense	2	14	32	46	58	76
Net Income	9	79	181	258	331	431

Balance Sheet						
Fiscal Year	2025 E	2026 E	2027 E	2028 E	2029 E	2030 E
Cash And Equivalents	31	129	316	584	915	1,337
Accounts Receivable	69	99	118	139	178	230
Inventory	2	3	3	4	5	6
Prepaid Exp.	9	14	18	23	29	38
Net Property, Plant & Equipment	36	41	47	55	63	74
Other Intangibles	62	62	62	62	62	62
Total Assets	297	435	653	956	1,341	1,836
Accounts Payable	29	36	37	45	54	66
Accrued Exp.	24	32	35	37	47	61
Unearned Revenue, Current	70	96	110	124	135	146
Total Current Liabilities	132	177	199	228	264	309
Long-Term Debt	124	124	124	124	124	124
Total Liabilities	327	373	395	423	460	504
Total Equity	(30)	62	258	533	882	1,333
Total Liabilities And Equity	297	435	653	956	1,341	1,837

Discounted Cash Flow Valuation						
Fiscal Year	2025E	2026E	2027E	2028E	2029E	2030E
Revenue	346	528	682	878	1,115	1,441
EBIT	25	106	226	317	403	520
NOPAT	21	90	192	269	342	442
D&A	6	9	12	15	20	27
Capex	(9)	(14)	(18)	(23)	(29)	(38)
Change in NWC	17	(11)	2	(2)	9	17
Unlevered FCF	35	74	188	260	342	448

Exit Multiple Approach	
Terminal year Revenue	1,441
Revenue multiple	2.5x
Terminal value in 2030	3,602
Present value of terminal value	1,859
Present value of stage 1 cash flows	730
Enterprise value (stage 1 + 2)	2,590
Enterprise value	2,590
Net debt	177
Equity value	2,413
Shares outstanding	165
Equity value per share	\$14.61

Equity Valuation using DCF





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Private Market Coverage

PRIVATE MARKETS OVERVIEW

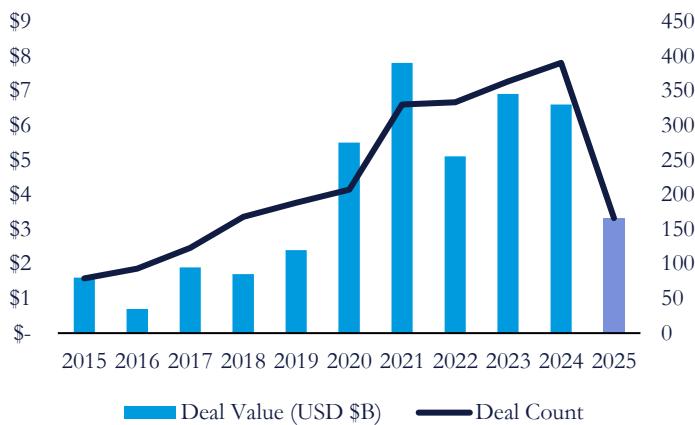
Historical Investment Trends

The private markets for space technology and manufacturing have seen explosive growth in the last decade, primarily driven by falling launch costs, defense contracts, and the commercialization of satellites. Cumulative private investment reached \$65.6B from 2015-2024 across 1,634 deals, of which 81% of funds came from VC. The sector stabilized post 2021-peak amid larger VC slowdowns but rebounded in 2024 with \$7.8B invested fueled by need for geospatial analytics and advanced infrastructure. Manufacturing subsectors such as spacecraft and launch vehicles captured 21% of total investment, reflecting high capital intensity for reusable tech and orbital factories.

Rise in Space Tech VC Funding

Venture capital investment in space technology is accelerating rapidly, with over \$3.3B deployed as of June 2025, and the sector is on track to hit a new all-time high. Investors are broadening their focus beyond rockets and launch vehicles toward downstream applications such as satellite networks, Earth observation, and defense-driven infrastructure. Heightened geopolitical tensions and national security priorities are de-risking the market, drawing sustain private capital into what was a niche domain. In an otherwise cautious VC landscape, space tech represents a bright spot, signaling that the industry may be entering a mature, commercially viable phase.

Space Tech VC Funding (USD \$B)



Emerging Trends & Risks

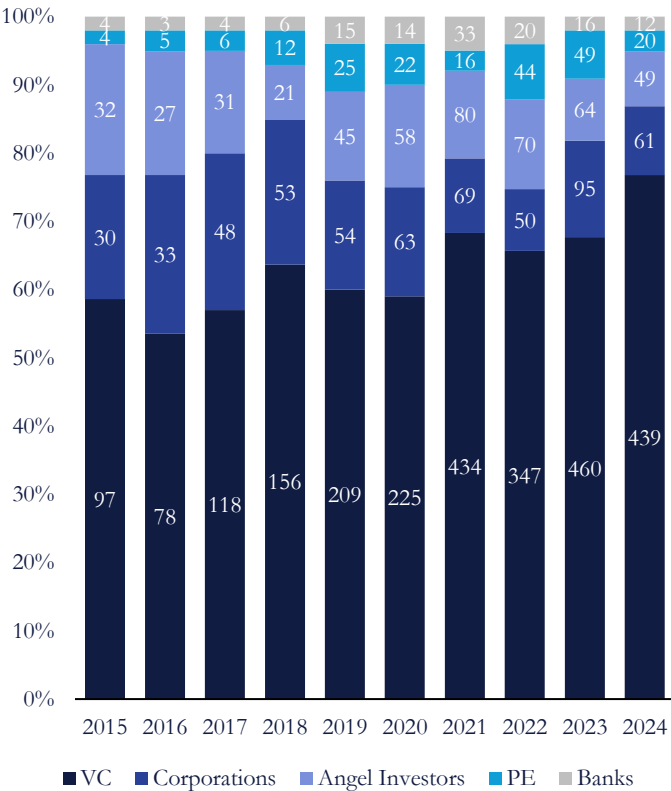
Trends:

- 1) Private space tech is accelerating as reusable launch systems drive down the cost and frequency of access to orbit, enabling both established players and startups to deploy more hardware.
- 2) The demand for satellite-based data and global connectivity is rising across defense, climate, logistics, and telecom sectors at unparalleled rates.
- 3) In-orbit infrastructure, like satellite servicing and debris management, are emerging, shifting space from one-off missions to ongoing services.

Risks:

- 1) The market still faces high technical execution risk as launch vehicles, constellations, and in-space servicing systems are complex, capital-intensive, and prone to delays or mission failures.
- 2) Regulatory pressure around orbital debris, traffic management, and spectrum access is increasing and may restrict operations.
- 3) Growing competition, especially in launch and small-satellite services, risks margin pressure unless companies differentiate clearly.

Investor Composition



Sources: BryceTech, Pitchbook, Bloomberg

FUTURE PROGRESS

Emerging Markets: Asteroid Mining

Asteroid mining is expected to yield 10x returns for early investors, with three commercial demos taking place over the next few years. The reason these missions are so sought after is because a single 500-meter M-class asteroid (metallic-type and rich in iron-nickel cores) holds ~\$3T in plutonium-group metals, split between ~100,000 tons of platinum and ~1 million tons of nickel and cobalt. This average looking rock single-handedly contains more platinum than has ever been mined on Earth. Outcomes of such mining could easily deliver 10-15x MOIC to seed investors who lock in off-take agreements today as the initial returns would disrupt supply chains and enable in-space refining for lower costs than terrestrial mining.

Asteroid Mining Rollout

2026  **ASTROFORGE**

AstroForge, a U.S. startup, will launch its Vestri mission to land on and refine metals from a metallic asteroid.

2027  **ispace**

Japanese company ispace, plans lunar payloads to prospect and extract helium-3, a rare isotope for fusion energy and quantum computing cryogenics.

2028  

UAE's MBR explorer mission is a multi-asteroid aircraft tour, culminating in a 2034 landing on the asteroid, Justitia, to map organics and metals.

Importance of Space Data Services

The market for space data services, valued at \$12B today, is projected to reach \$30-45B by 2030, with private startups raising \$3.2B in 2025 alone for AI-driven analytics. In logistics, energy, and heavy infrastructure, affordable satellite imagery now enables predictive maintenance and risk mitigation at new levels: global fleets cut fuel costs by up to 15%, supply-chain digital twins anticipate weather or geopolitical disruptions with 90%+ accuracy, and just-in-time manufacturing trims inventory holding costs by 20-30%. Perhaps most critically, emergency response for dams, pipelines, and ports has shrunk from days to hours.

Government Demand for Space Tech

The convergence of government funding and private investment is reshaping global space development. NASA's Artemis program exemplifies this synergy: SpaceX holds \$4.4B in Starship HLS contracts, while Blue Origin secured \$3.4B for Blue Moon, with \$20B+ in total NASA awards projected by 2030. In defense, NATO's 2025 Commercial Space Strategy and U.S. Space Force awards drive 15% YoY private funding growth, with 78% of the 2024 space economy now commercial. Dual-use tech, from Starship refueling to Blue Origin engines, multiplies impact across missions, and for industrials, this means stable, high-return revenue from propulsion, materials, and analytics.

Vertical Integration

The newfound shift towards vertical integration is enabling end-to-end control for players like SpaceX and ULA, reducing costs by 20-30% and accelerating commercialization, drawing \$5.5B+ in funding for 112 start-ups.

Upstream

Development, Manufacturing & Testing:

- Uncrewed/crewed vehicles
- Satellite bus platforms/payloads
- Orbital station infrastructure

Midstream

Operations:

- Spaceport and launch ops.
- Astronaut training/monitoring
- Payload operations and ground stations

Downstream

Services:

- Satellite communication
- Remote sensing analytics
- IoT infrastructure control
- Location-based services



DEAL ENVIRONMENT

Northrop Investment in Firefly

5/29/2025

\$50 million

Northrop Grumman Corporation

Firefly Aerospace

Northrop has made a \$50 million investment into Firefly Aerospace to advance the development of the companies' co-developed medium launch vehicle: Eclipse. Firefly had previously raised \$175 million for a Series D round and was valued at over \$2 billion in November 2024. Filling a void in an underserved market, Eclipse will launch up to 16,300 kg of cargo to low Earth orbit or 3,200 kg of cargo to geosynchronous transfer orbit, expected to launch in 2026 from Virginia. This investment provides a strategic move for Northrop, strengthening their medium-lift offering (a segment gap for the company), while getting exposure to a high-growth commercial launch partner. While being a minority investment, this deal signals sizeable market confidence in Firefly's operations and backlog. Firefly's focus now shifts heavily to executing on the Eclipse development timeline and transitioning to revenue-generating launches from development stage. This deal holds considerable weight in the maturing space-tech private market as it signifies a sizeable investment from an established aerospace contractor taking a strategic stake in an emerging launch providers with the industry shifting focus toward next-generation launch systems.



True Anomaly Series C Funding

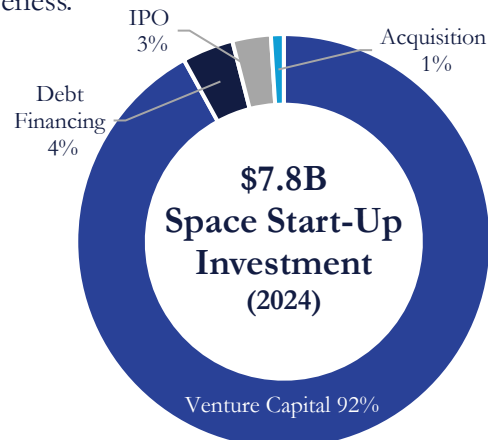
4/30/2025

\$260 million

Accel and other VC firms

True Anomaly

True Anomaly is a US-based space-defense startup focused on building autonomous orbital systems designed to protect national assets in space. Their core platform is Jackal, a military-class spacecraft which enables real-time monitoring, maneuvering, and mission assurance, while their Mosaic software processes telemetry and predicts failures before they occur. As satellite constellations grow exponentially, reliability and security have become critical infrastructure, and True Anomaly positions itself as the “operational layer” that enables safe, controlled space environments through AI-driven situational awareness.



Anomaly announced a \$260 million Series C round April 2025, led by Accel, with participation from Meritech Capital, Menlo Ventures, and more, and debt financing from Stifel Bank. This raise is one of the largest in the defense-space sector and reflects investor belief that the commercialization of orbit now requires enterprise-grade security, persistent tracking capabilities, and scalable satellite servicing infrastructure. This deal matters because it signals a shift in the private market toward mission-critical space infrastructure rather than speculative exploration. Companies like True Anomaly have become strategic assets not just for commercial customers, but for national defense and intelligence agencies as nations race to secure orbital dominance.

FUTURE AMIDST VOLATILITY

Talent Arbitrage

The commercial space industry has decisively overtaken traditional government programs and legacy primes as the destination of choice for elite aerospace engineering talent. According to Deloitte's 2025 Aerospace & Defense Outlook and PwC's 2025 Space Industry Trends report, approximately 70–75% of new aerospace PhDs and top-tier graduates from institutions such as MIT, Caltech, and Purdue now join private-sector firms rather than NASA or large defense contractors, a complete reversal from a decade ago. This shift is driven by compensation and mission velocity: total cash-and-equity packages at leading private space companies routinely exceed those at the primes by 30–50% across experience bands (median aerospace engineer salary is \$134,830, while private space equivalents are around \$150k at entry level). For equity investors, this migration creates a powerful and widening moat, making the best intellectual capital available to highest-conviction private teams at valuations that remain a fraction of public primes. The result is an accelerating transfer of proprietary know-how from the public sphere to venture-backed entities, mirroring talent dynamics that proved decisive in early Tesla and SpaceX success.

Portfolio Convexity: Strength in Crises

When markets crash or wars start, most alternatives collapse; however, space tech often benefits. The 2008 financial crisis catalyzed the GPS commercialization boom, the COVID pandemic accelerated Starlink adoption, and the 2022 invasion of Ukraine generated \$4B+ of DoD contracts for resilient communications infrastructure and satellite systems deployment. Future tensions in Taiwan and the Middle East are expected to drive similar strategic demand spikes. Space remains one of the only private asset classes with a positive beta to geopolitical risk. In other words, it actively hedges against Earth-bound volatility. A singular event combined with a 3x carrying position can convert into 8–12x returns almost overnight.

	Normal / Bear Market	Geopolitical Crisis
Most Alternative Assets	(30%) to (70%)	(50%) to (90%)
Space Tech	(20%) to +20%	+150% to +400%

The Incoming 2030 Liquidity Supercycle

Most market participants remain fixated on the near-term IPO windows, yet the far more predictable exit pathway will be strategic consolidation. By 2030, more than 50 public A&D primes and large defense contractors must close aggregate revenue gaps exceeding \$36 billion in their space segments to meet stated 2035 targets, creating a classic forced-buyer dynamic that drives aggressive acquisition mode. Lockheed Martin, Northrop Grumman, RTX, and Airbus Defense alone face a combined \$36 billion organic shortfall that cannot be filled internally at acceptable returns. In this environment, any private company demonstrating profitable scale in launch, satellites, or in-space servicing rapidly becomes a strategic necessity rather than an opportunistic buy. Private-to-private and private-to-public roll-ups will therefore dominate the next decade's liquidity channel, pulling forward transaction timelines and inflating strategic premiums on differentiated assets. Leading private investors already capitalize on this dynamic in 2025, securing top-tier space companies at pre-money revenue multiples of 15-20x.

Consolidator	2035 Space Revenue Target	Current Organic Run-Rate	Required Bolt-On Revenue	Estimated Acquisition Budget
Lockheed Martin	\$20B	\$9B	\$11B	\$50B
Northrop Grumman	\$16B	\$7B	\$9B	\$40B
RTX	\$14B	\$6B	\$8B	\$35B
Airbus Defence	\$14B	\$6B	\$8B	\$30B
Total	--	--	\$36B	\$155B Acquisition Capacity

Closing Remarks

Space technology remains an early-stage sector, and near-term valuations will naturally exhibit volatility. Unlike traditional industrial verticals, it is not yet a foundational part of the global economy, although it is positioned for outsized growth as demand for off-planet capabilities accelerates. As terrestrial resources tighten, the long-term opportunity in space-based exploration, resource extraction, and orbital infrastructure expands materially. Emerging propulsion technologies, including solar sails and advanced nuclear systems, make deeper space operations scientifically credible within this century. Government investment will continue to anchor early development, but public-market investors are poised to participate in the substantial value creation that will emerge as humanity begins to access and commercialize resources beyond Earth.



The 2025
Fall Industrials
Team thanks you
for reading and following
along with our report. These
are exciting times for space companies,
and public markets and private markets alike
have started to bring incremental capital to the table
for the smartest engineers and astrophysicists to innovate beyond any previous capacity. Humanity's ability to innovate has always depended on access to new frontiers, and Earth's natural resources and physical constraints are increasingly limiting. Space offers that next frontier. Whether through advanced telecommunications, off-planet resources, orbital manufacturing, data centers, or commercial spaceflight, we expect space technology to become a major engine of global growth. The companies building this ecosystem today are shaping what will become one of the most transformative markets of the next century.

The future is moving upward, and it's better to be early than late.

Meet the Team



Miles Rucker is a Finance undergraduate student with a minor in Philosophy. Miles bring experience from Wellington Management, in their Americas Institutional Group. Miles currently serves as the Director of PEVC Case, the President of Scholars of Finance, and Athlete on the Northeastern Alpine Ski Team.

Miles is currently on co-op at Wellington Management and is looking to pursue another buy-side role in the following cycle.



Owen Romano is an undergraduate student pursuing a degree with a dual concentration in Finance & Accounting. Owen is currently working in real estate development at Ionic Development company, with prior experience in the real estate space at BXP, Inc. This is Owen's fourth semester in Bull & Bear Research, and he also is working towards co-founding a credit analysis club at Northeastern.

Upon completion of his current co-op, Owen is looking to pursue full-time opportunities in real estate lending to further his knowledge of real estate financial markets at large.



Zachary Cohen is a Junior pursuing a combined degree in Finance and Political Science. Zachary brings experience from his role on the M&A Insights team at McKinsey & Co. He also has experience working in private investing at Raptor Group. Zachary is also an Analyst at NUImpact, Northeastern University's student impact investment fund.

Zachary is currently on co-op at McKinsey & Company on the M&A Insights and is looking to pursue a role in consulting, buy-side investing, or Investment Banking in the following cycle.



Rohan Kalaria is an undergraduate student pursuing a degree in Finance. Rohan brings experience from his role as an Investment Banking Analyst at Concordia Capital. Rohan is also an involved member in Scholars of Finance and the Professional Business Fraternity, Delta Sigma Pi.

Rohan is looking to pursue his first co-op in Investment Banking or Private Equity to further explore the deal process and refine his technical skills.



Gonçalo Soares is a second-year pursuing a degree in Finance. Gonçalo brings experience from Merck & Co Inc in their Finance and Treasury Groups. Gonçalo also currently serves as a Junior Analyst at PEVC Case.

Gonçalo is looking to pursue his first co-op and summer internship in Investment Banking or Private Equity to gain firsthand experience working on live-deals and learning the intricacies of valuation.